

Bridging the Gap:

From Innovation to Patient Impact

Ethics, Evidence, Regulation and Responsible Medical Device Translation

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2 devices

2 patients

6 stages

5 decisions

2 votes

Authentic device photographs • no patients shown



FoGO

Freezing-of-gait wearable • alpha prototype



SwaKnee

Pulsed electromagnetic field (PEMF) knee system

Six stages • six patient questions • two composite patients



Need



Evidence



People



Permission

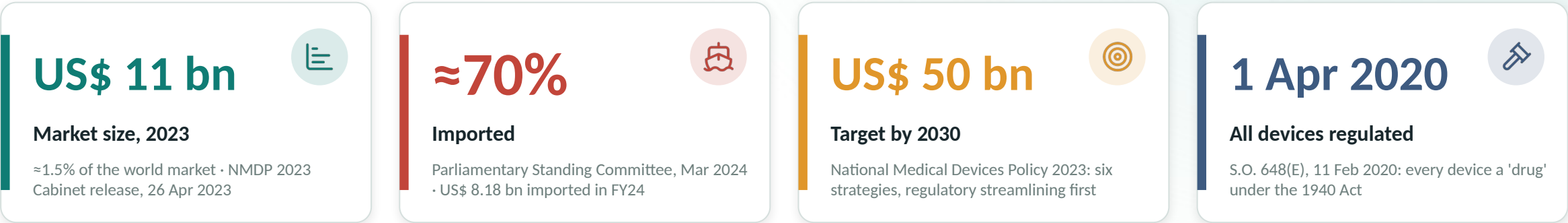


Access



Safety

India's medical device moment



One device's regulatory journey under the Medical Devices Rules, 2017 (MDR-2017)



Regulation is a sequence of permissions, each answering a different patient question: the same six stages structure this session.

My interests, stated first

Interests in the two case devices



FoGO • Ahilaya Biomedicals Pvt Ltd

Freezing-of-gait wearable • incubated at KIIT TBI

Alpha prototype • MD-13



Founder and principal investigator



Provisional patent 202531119165 and a design application filed



₹25.5 lakh in grants: Startup Odisha • DST NIDHI PRAYAS • Startup India Seed Fund



BIRAC Biotechnology Ignition Grant proposal under review (BIG: up to ₹50 lakh over 18 months)



SwaKnee • Swayogya Rehab Solutions

Pulsed electromagnetic field (PEMF) knee system



[state role and any equity, royalty or salary interest]

Three ground rules for today

1

Vote before you hear my view

Record your intuition first; we compare it with the closing vote on the same options.

2

Challenge every claim, including mine

Ask of each claim: what is the evidence, who measured it, and who paid for it?

3

Protect identities

No names or photographs of patients from your own practice; personas here are composites.

A conflict of interest is a set of circumstances that creates a risk that judgement about a primary interest will be unduly influenced by a secondary interest (Institute of Medicine, 2009): a risk, not an accusation. Test everything I say by whether a conflict would be reasonably perceived (University of Cambridge policy).

OPENING VOTE

Would you let a patient use it?

Three facts about one device (not named yet)



A working prototype



Promising results on public datasets



A test licence granted

Record the split on the flipchart

1 = ____

2 = ____

3 = ____

Three options (one, two or three fingers)

1

Routine care

Suitable for a clinician to recommend to patients now

2

Research study

Suitable for a study with ethics approval and consent

3

Not yet

I need more information before either

Hold up one, two or three fingers, then tell your neighbour why. We vote again at the end with exactly the same options.

Three teaching traditions, one Indian patient



Stanford Biodesign

NEED

- Identify → Invent → Implement: start from a validated need, not a technology
- Need Statements (Student Guide, 2022): solution-neutral, screened before inventing
- Principled Decision-Making (2022): write the team's ethical principles early
- India: Stanford-India Biodesign since 2007 (32 fellows, 14 technologies, 13 companies), now the School of International Biodesign at AIIMS; an AIIMS emergency-department study filtered 100 needs to 10 (Chaturvedi 2015)



Harvard research ethics

PEOPLE

- HMS Center for Bioethics: case-based teaching of research ethics
- MRCT Center (Brigham and Harvard): Diversity guidance (2020), Accessibility by Design toolkit (2023), plain-language glossary, post-trial access for devices (2025)
- Petrie-Flom Center: Diagnosing in the Home; ethics of remote device monitoring (Cohen, Gerke & Kramer 2020)
- HMS conflict-of-interest policy (2010): disclose, then manage



Cambridge design and translation

ACCESS · SAFETY

- Engineering Design Centre Inclusive Design Toolkit and Exclusion Calculator: who is excluded by a product's demands?
- Engineering Better Care (2017; working group chaired by Prof John Clarkson): people, systems, design and risk, with a 'Sustain' phase
- Institute for Biomedical Innovation (announced January 2026): from lab prototype to trial-ready device
- Cambridge Judge CCHLE: making proven solutions affordable

None of these institutions has evaluated or endorsed FoGO or SwaKnee. The Biodesign textbook is Stanford content published by Cambridge University Press.

Applied to Indian rules and Indian patients



CDSO · MDR-2017

Permission: Rule 4 classes A–D; licences MD-5 / MD-9; test licence MD-13



ICMR 2017

People: National Ethical Guidelines; ethics committees registered under NDCT Rules 2019



MvPI · IPC

Safety: adverse-event reporting since 6 Jul 2015; 174 monitoring centres



NPPA · DPCO 2013

Access: ceiling prices on stents and knee implants (2017); trade-margin caps (2021)



BIS

Standards: Indian Standards; Essential Principles checklist (Class A self-certified)



DPDP Rules 2025

Data: personal health data from apps and cloud; phased commencement

Two people we will follow

Ramesh and Kamala are composites, not real patients, built from situations common in Odisha clinics. Every decision today is made for one of them: when you vote, vote for them.



Ramesh, 71

Parkinson's disease with freezing of gait

≈7.7 lakh

people living with Parkinson's disease in India, 2019
India State-Level Disease Burden Initiative (GBD 2019), as cited in the FoGO project material

- ⌚ Parkinson's disease for eight years
- 🚶 Freezes in doorways and when turning
- ⚠️ Has fallen twice this year
- 📱 Lives with his daughter, who works days and lends him her smartphone

Device in this story: **FoGO, freezing-of-gait wearable**

Alpha prototype • MD-13



Kamala, 64

Knee osteoarthritis in both knees

28.7%

knee osteoarthritis in a five-site community survey (n = 5,000)
Pal et al., Indian J Orthop 2016 (one study, not national)

- 🦴 Osteoarthritis in both knees
- 📍 Lives 30 km from the clinic at the district hospital
- 🚗 Travels by shared auto; someone must accompany her
- 💰 Household pays out of pocket and loses wages on clinic days

Device in this story: **SwaKnee, PEMF knee system**

Product • 45-min daily sessions

Every stage answers a patient question



The six dots at the top of later slides show which stage we are in. Stages are not strictly sequential: Engineering Better Care's 'Sustain' phase means access and safety are designed in from the start, not added after launch.

01

PART 1 • NEED AND EVIDENCE

What are we promising, and what has been shown?

Two devices, two patients. First the wording of the need and the claim; then what each prototype has demonstrated, reported, planned or not yet established.



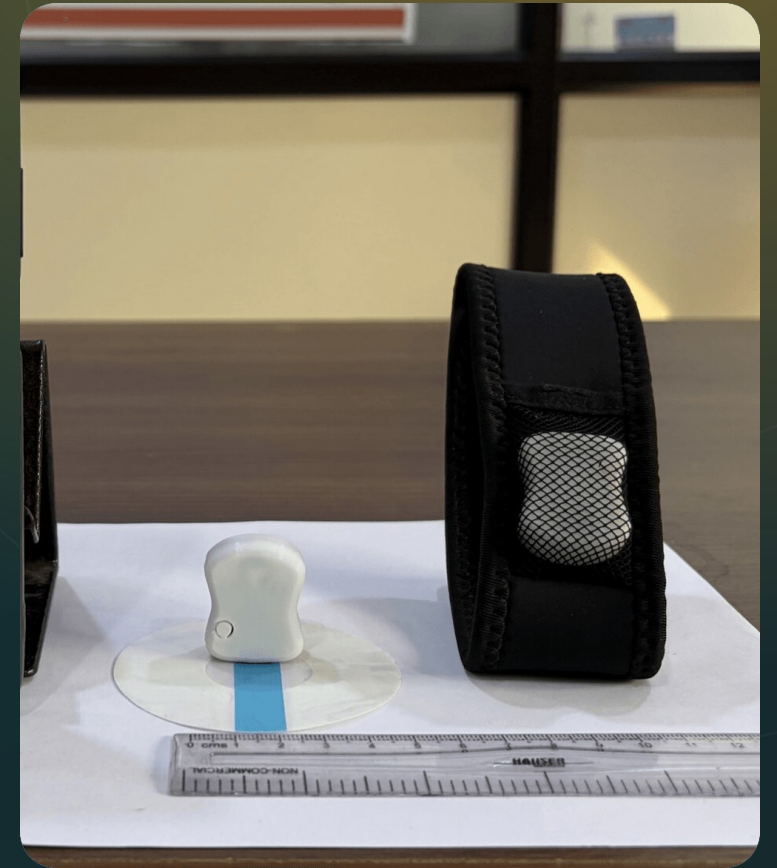
A solution-neutral need, and a claim ladder that sets the evidence bar



Is it a medical device in India? The intended purpose decides, not the technology



FoGO and SwaKnee on the evidence ladder, with a decision to vote on



FoGO alpha prototype (V3, 2025): phone app, chest module on an adhesive patch, ankle module

Wording sets the evidence bar



Stanford Biodesign need statement

A way to [address the problem] in [a specific population] in order to [achieve a desired outcome]

Ramesh: a way to **reduce freezing-related immobility and falls** in **people with Parkinson's disease living at home**, in order to **preserve safe, independent walking**.

No sensor, no vibration, no app: the need stays solution-neutral.

Your turn (60 s): rewrite Kamala's need without naming any device.



Why the words matter. Under MDR-2017 the intended use and the claims decide whether a product is a medical device and which class (A-D) it falls in. CDSCO's software guidance (21 Jul 2026) grades risk by how far the output drives clinical decisions and how serious the condition is.

INFORMATION

"Shows your walking patterns"

Evidence needed

- Bench accuracy
- Usability

Permission in India

- No medical purpose: outside MDR-2017
- Real purpose decides, not the label

MONITORING

"Tracks your gait for your neurologist"

Evidence needed

- Analytical validation of the measure
- Software V&V, cybersecurity
- Usability in Indian clinics

Permission in India

- Medical purpose: a device
- SaMD class by MDSW guidance 2026
- Licence MD-5 (State) or MD-9 (Central)

DETECTION

"Detects freezing of gait in Parkinson's"

Evidence needed

- Sensitivity, specificity vs video-labelled freezes
- False alarms per hour, latency
- Homes, turns, doorways, walking aids

Permission in India

- Test units: MD-12 → MD-13
- Clinical investigation: MD-22 → MD-23
- Registered ethics committee, CTRI
- Then the licence

TREATMENT

"Reduces freezing and falls"

Evidence needed

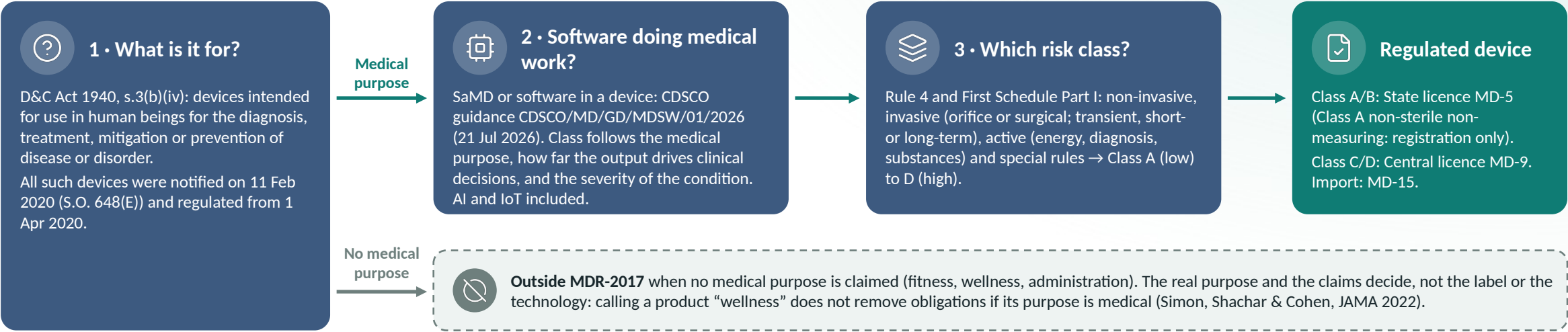
- Prospective controlled trial: falls, walking, confidence
- Durability and safety over time
- Fair performance across groups

Permission in India

- All of the detection rung
- Claims control: UCMPMD 2024, DMRA 1954, ASCI
- Label per Rule 44; vigilance after launch

Stronger claim → more evidence, more permissions, closer scrutiny

Is it a medical device in India?



FoGO **Device**

- Claim: detects freezing of gait and cues the user
- Software drives the cue: SaMD rules apply
- Class: developer proposes B; say “likely B or C” until CDSCO confirms

SwaKnee **Device**

- Claim: pulsed electromagnetic field for knee osteoarthritis pain
- Active therapeutic device delivering energy
- Class: likely B under the First Schedule; verify on CDSCO lists

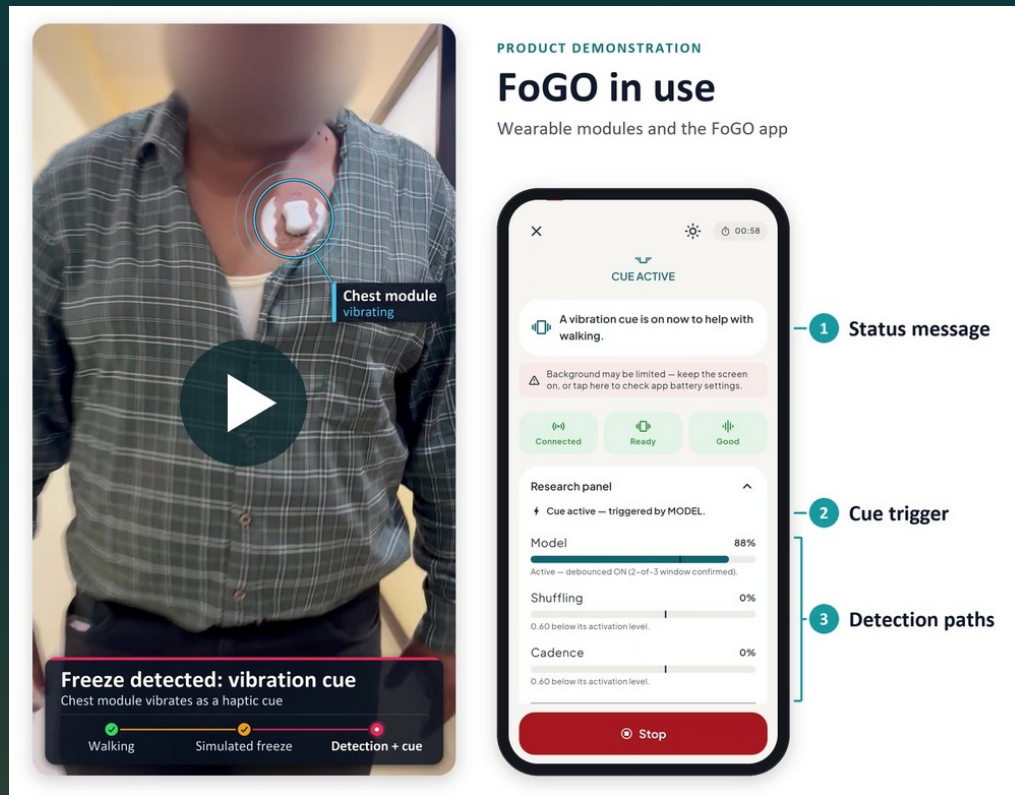
Fitness band **Not a device**

- Claim: counts steps, “stay active”
- No disease named, no diagnosis
- Adds “detects arrhythmia”? Then it is a device

Hospital app **Not a device**

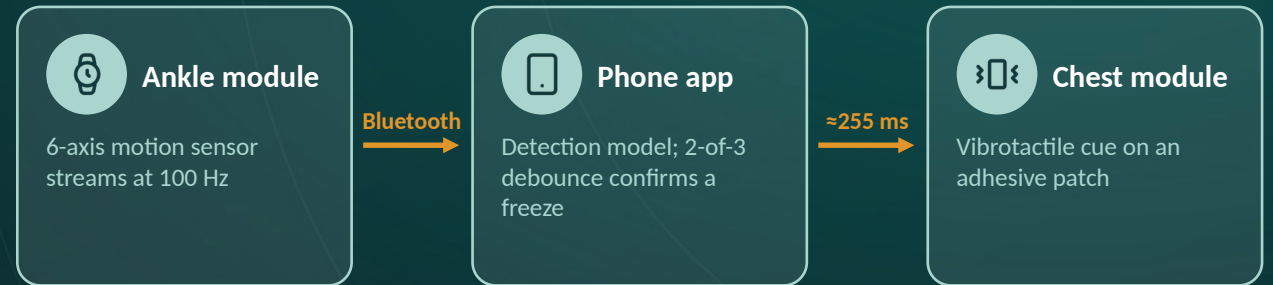
- Appointments, queues, billing
- No medical purpose in the output
- Adds triage or dosing advice? Re-check

What did the prototype show?



12 s staged demonstration • volunteer • simulated freeze • face obscured • not clinical footage

Watch, then: what did you see? What did you infer?



Detection-to-cue latency ≈ 255 ms and 100 Hz sampling are project-reported bench values, not clinical measurements.

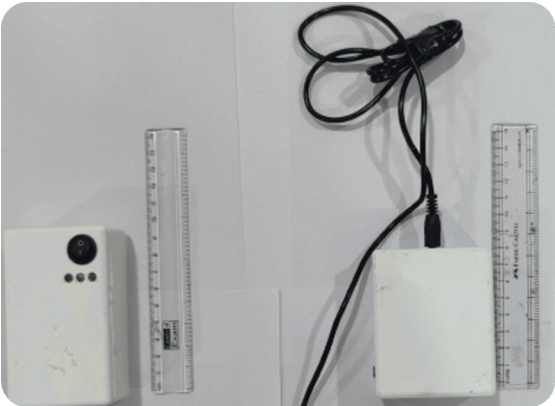
Seen (demonstrated)

- ✓ Stop sensed by the ankle module
- ✓ App thresholds crossed; flagged as a freeze
- ✓ Chest module vibrates on cue

Not yet shown

- ✗ Real freezes in people with Parkinson's
- ✗ A freeze told from a voluntary stop or a doorway pause
- ✗ Many users, homes, turns, walking aids
- ✗ Tolerability of repeated cues
- ✗ Fewer falls or more walking

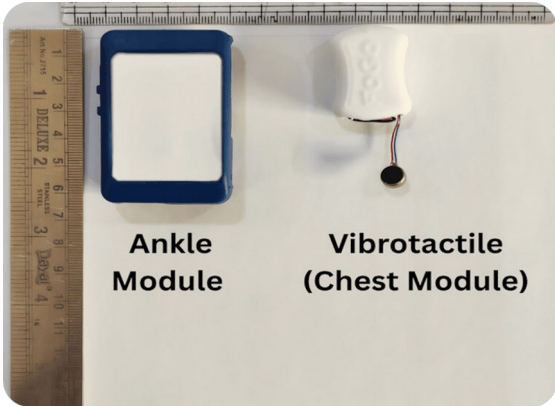
Three iterations, three kinds of proof



V1 • Jan 2024 • boxed electronics, wired motor

Demonstrated

Bench function: sensing electronics and a vibration motor working from a box



V2 • 2024-25 • two wearable modules

Demonstrated

Wearable form: ankle sensor case and a small chest vibrotactile unit



V3 • Jun 2025 • alpha system with app

Demonstrated

Integrated system: ankle strap, adhesive chest patch, phone app; the staged demo



EMC pre-compliance • Jul 2026

Pre-compliance, not certified

IEC 60601-1-2 pre-compliance run; formal testing for the dossier still to come

None of these is clinical evidence: each prototype answered an engineering question. The MD-13 test licence allows 25 units for investigation, test and demonstration; they cannot be sold.



FoGO has climbed two rungs, not five

5

Safe and fair over time?
Long-term safety; fair performance across users, aids and homes

Not established

4

Helps in daily life?
Fewer falls, more walking, confidence. No cueing trial has yet shown fewer falls

Not established

3

Detects real freezes in people?
Prospective proof-of-concept with AIIMS Bhubaneswar advisers; BIRAC BIG proposal under review; not yet funded, approved or permitted

Planned

2

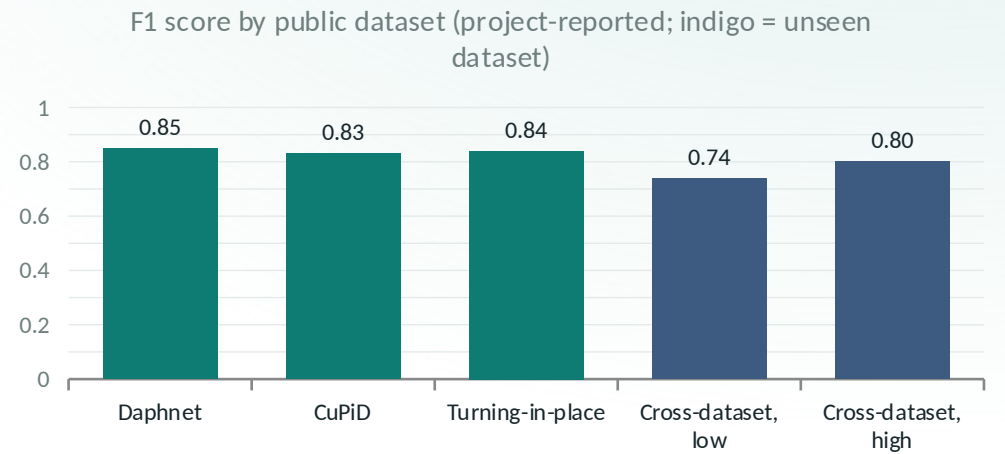
Detects on recorded data?
F1 0.83–0.85 subject-wise on Daphnet, CuPiD and Turning-in-place; 0.74–0.80 on an unseen dataset; project-reported, not peer-reviewed

Reported, preliminary

1

Works as a system?
Staged demo: a volunteer's simulated freeze sensed, flagged and cued; bench study of five cue sites (chest most sensitive)

Demonstrated



0.15–0.18

False alarms per minute

Subject-wise; ≈0.21–0.27/min cross-dataset (project chart)

n = 10

Daphnet benchmark

Lab, 8 froze, 237 video-labelled events

Pre-IDEAL (bench, datasets) → IDEAL 1–2a (first in humans) → 2b–4 (comparison, long-term)

The rungs are questions, not a compulsory sequence. Which rung matters most before Ramesh uses it at home?

Home pilot for Ramesh next month?

A hypothetical scenario built on real FoGO facts



F1 0.83–0.85 on public data



Ten users, alone at home



Ramesh fell twice; asks to join

Vote A, B or C. Then pairs, 60 seconds: if B, which two conditions?



Proceed



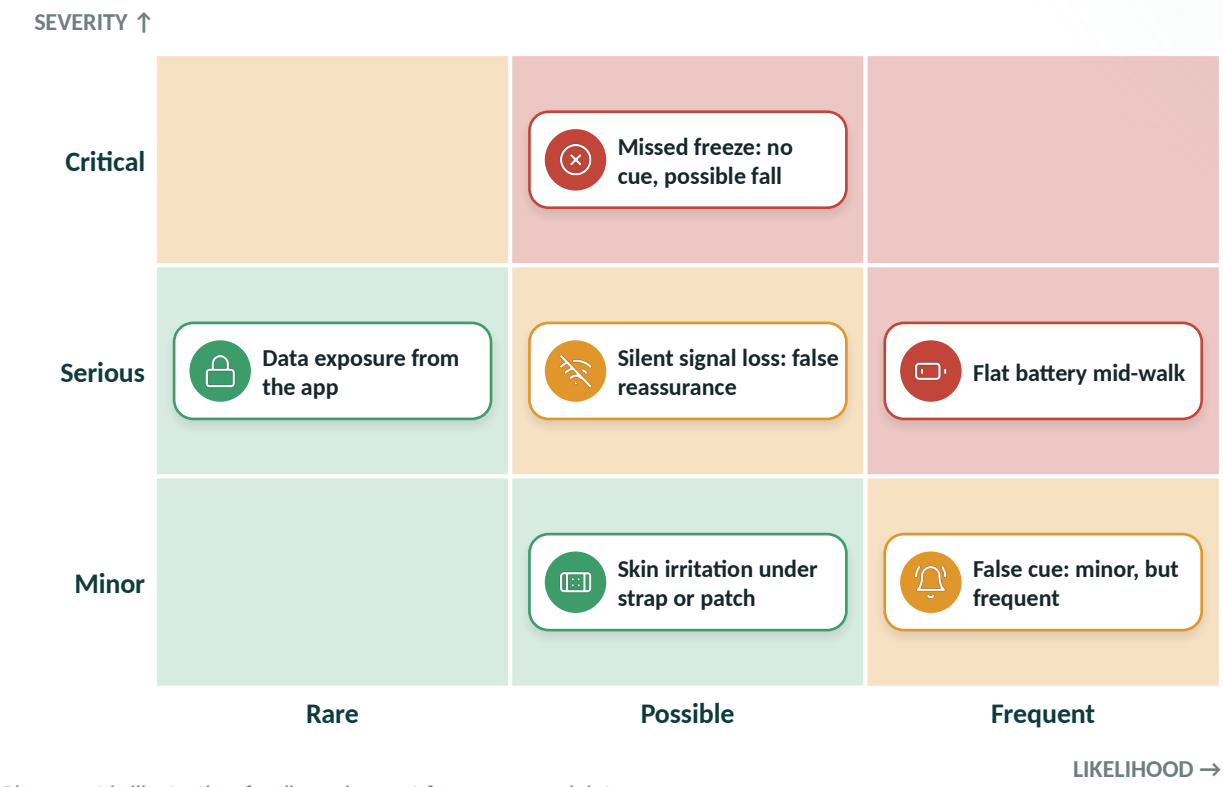
Proceed with conditions

Name them



Not yet

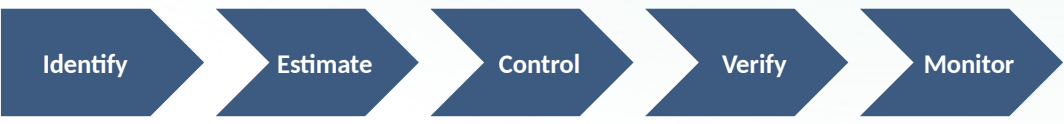
Rank failures by harm and likelihood



Placement is illustrative, for discussion; not from measured data.

9-16

unnecessary vibrations per walking hour
if the reported 0.15-0.27 false alarms per minute held at home. An extrapolation from dataset rates, not a measured home rate. Would Ramesh keep wearing it?



ISO 14971:2019 risk management • IEC 62366-1 usability engineering • ask how each control will be shown to work

Controls to test first

- Visible and tactile loss-of-monitoring alert
- Cue-intensity limit (within the bench-tested range) and an easy stop
- Performance during turns, in doorways and with walking aids
- Low-battery warning before a walk
- Encryption and short data retention (DPDP Rules 2025)

Which control would you test first, and how would you show that it works?

A device is also a daily routine



1 Fit the applicator

20 s company demonstration of use · fit, switch on, rest · shows no clinical response · no faces

Reported: company material

≈34 hours

45 min a day × 45 days = 2,025 minutes

*of Kamala's time, before travel, fitting, charging and help from family
(leaflet schedule: an illustration, not a prescription)*

1

Fit the applicator around the knee
Strap cuff; connect the cable to the controller

2

Switch on the controller
Power button; status light; mains or charged battery

3

Rest for the 45-minute session
Seated or lying; every day

4

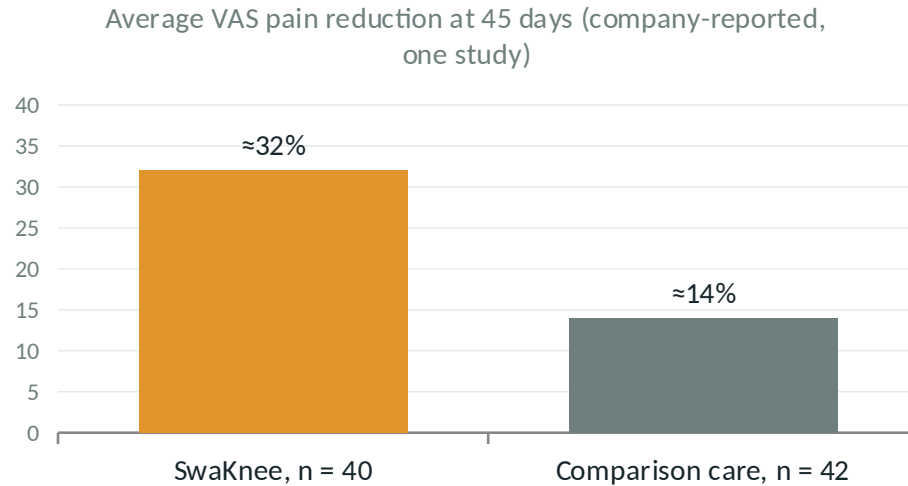
Repeat for 45 days
Charge, store, travel for follow-up; family helps when it slips



Controller and knee applicator

Who helps Kamala when the cuff slips, the controller shows an error, or her knee feels warm?

Which claim can this evidence carry?



Patient's yardstick (MCII, knee OA): -19.9 mm, or -40.8%, on a 100 mm pain scale before a person calls the change important (Tubach 2005). An average fall of 32% does not say how many people reached it.



Industry-sponsored drug and device studies reach favourable conclusions more often: RR 1.34 (Lundh et al., Cochrane 2017).

One of you defends claim 1; another challenges claim 3.

1

"Less pain on average in one 45-day company study"

Reported (company)

Acceptable if labelled company-reported, not peer-reviewed

2

"Clinically proven"

Not established

Needs independent, registered, sham-controlled trials with consistent results

3

"Regrows cartilage"

Not measured

No structural outcome measured; no human evidence of regeneration found



What the public summary does not report

- How groups were allocated
- Blinding or a sham control
- Prospective CTRI registration
- How missing data were handled
- Between-group difference with its confidence interval
- Adverse events
- Durability beyond 45 days
- Device and firmware version studied

What independent evidence says about cueing and PEMF



FoGO domain • detecting and cueing freezing of gait



Daphnet benchmark (Bächlin 2010): 10 patients in a laboratory, 8 froze, 237 video-labelled freezes

Benchmark, n = 10



Silva de Lima 2017, systematic review: sensitivity 73–100%, specificity 67–100%, mostly laboratory studies

Lab-dominant



RESCUE trial (n = 153, home cueing): small gait gains and lower freezing severity in freezers; fall counts not measured

Falls not measured



Ginis 2018 narrative review: no trial found showing fewer falls from cueing, continuous or on-demand

Fewer falls: not shown



FoGO: project-reported F1 on public datasets; prospective study planned, not done

Reported, preliminary



SwaKnee domain • PEMF for knee osteoarthritis



Cochrane 2013 (9 trials, 636 adults, all OA sites and field types): pain probably improves ≈15/100 more than sham; function uncertain

Moderate: pain only



Chen 2019 (knee, 8 RCTs): no pain advantage vs placebo. **Yang 2020 (all sites, 16 RCTs):** an advantage

Reviews disagree



OARSI 2019 guideline: electromagnetic therapy strongly recommended against (low-quality evidence)

Strongly against



NICE NG226 (2022): do not offer listed electrotherapies, insufficient evidence of benefit (list does not name PEMF)

Insufficient evidence

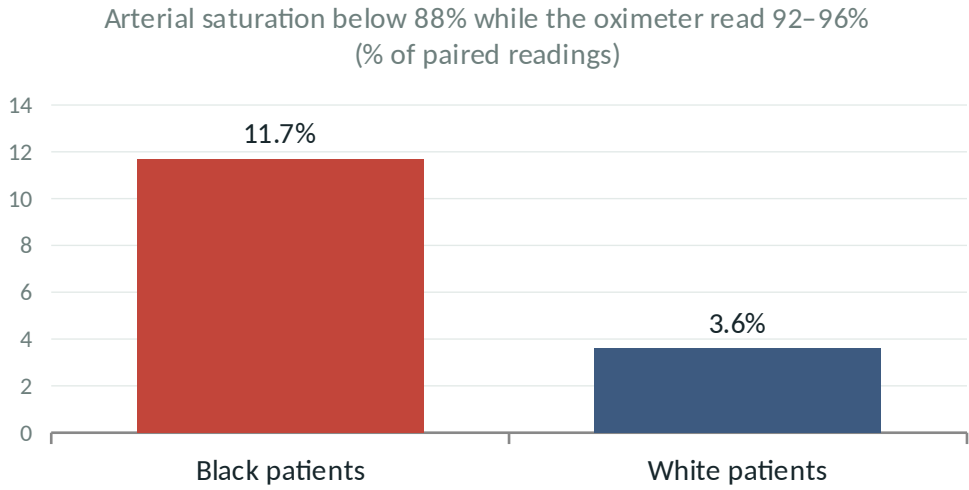


Cartilage regrowth: no human evidence found in any review; pooled outcomes are pain, stiffness, function

Not established

“Not found” in our searches is not proof of absence. Cochrane 2013 pooled all osteoarthritis sites and field types; NICE NG226's list does not name PEMF explicitly.

An average can hide a patient



Sjoding et al., NEJM 2020. 88 of 749 vs 99 of 2,778 paired measurements (95% CI 8.5–16.0 vs 2.7–4.7), University of Michigan cohort. Observational; race recorded, pigmentation not measured. It shows a device that looked accurate on average missing dangerous hypoxaemia three times as often in one group; it does not justify a race-based correction.

Regulator response

US FDA draft guidance, 7 Jan 2025: clinical testing of pulse oximeters across a diverse range of skin tones, assessed on the Monk Skin Tone scale.

Who is missing from our data?

Ramesh • FoGO

- People using walking aids
- Cognitive fluctuation
- Crowded homes, floor-level living
- Saris or dhotis over the sensor
- Public datasets with few or no Indian participants

Kamala • SwaKnee

- Women
- People with obesity
- Manual and agricultural workers
- People with other illnesses and medicines
- Rural users far from follow-up

MRCT Center diversity guidance (2020): diversity includes comorbidities, concurrent medicines and environment; report subgroups. **Stanford Biodesign health equity:** equity is decided at identify, invent and implement.

Take two answers for each patient. Who would our data miss?

02

PART 2 • PEOPLE

Can they say no, and who gains if they say yes?

Consent that is free and understood, and interests that are declared and managed: the people side of translation, under Indian rules.



Decision 2 of 5: Ramesh's neurologist is also the inventor



What valid consent requires in India: ICMR 2017, NDCT 2019, Helsinki 2024



Conflicts of interest: disclose, then manage; the rules in India and abroad

WHAT A PARTICIPANT WOULD ASK



“Can I freely say no?”

Helsinki 2024, para 27: in a dependent relationship, consent must be sought by someone independent of it



“Do I understand what I am agreeing to?”

ICMR 2017, s. 5: a language the participant understands; an impartial witness if they cannot read



“Who gains if I say yes?”

ICMR 2017: interests disclosed to the ethics committee; IMC 2002 cl. 6.8 and UCMPMD 2024 for doctors and companies

His neurologist is the inventor

A fictional vignette built on real consent rules; it describes neither project's actual procedure



“Will saying no change my care?”



Consent form: English only



Only neurologist within 100 km



Twist, if time allows: the clinic also wants to film him for a talk. Does that change your vote?

Vote A, B or C. Then 60 seconds of role-play: Ramesh, the recruiting neurologist, an observer. Observer: what would make refusal easier?

Three options



Recruit as planned

The neurologist takes consent himself, this week



Proceed with safeguards

Which ones? Name two



Not yet

Fix the consent process before anyone is approached

Independent consent-taker

Odia or Hindi, teach-back

Interest disclosed to the EC

What valid consent requires in India

Eight requirements of valid consent • ICMR 2017 ss. 5–7 • NDCT Rules 2019

- 1 **Information, key facts first**
Purpose, procedures, risks, benefits, alternatives; the US Common Rule (45 CFR 46.116) starts with key information
- 2 **A language the participant understands**
Odia or Hindi, not English only; plain words; teach-back (MRCT plain-language glossary)
- 3 **An impartial witness if the participant cannot read**
A literate witness unconnected to the research signs with the participant (ICMR 2017, s. 5)
- 4 **Legally authorised representative where capacity is impaired**
Fluctuating cognition, older and dependent people: LAR consent plus assent (ICMR 2017, s. 6)
- 5 **Audio-visual recording where the rules require it**
NDCT Rules: vulnerable participants in new-drug trials; ICMR allows AV documentation, witness in frame
- 6 **Re-consent when anything material changes**
New risks, a new software version or protocol: ethics-committee amendment, then the participant again
- 7 **Withdrawal at any time, without penalty**
Refusal or withdrawal must not change care (ICMR 2017; CIOMS 2016 Guideline 9)
- 8 **Compensation for research-related injury**
Free medical management and compensation; insurance is on the CDSCO Form MD-22 checklist

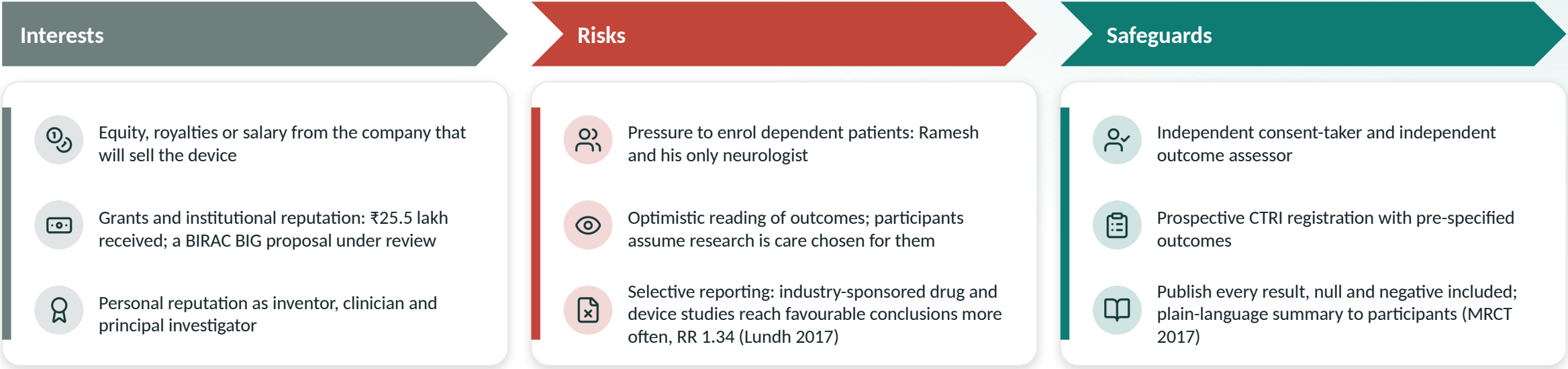
Before the first participant is approached

-  **Ethics committee registered with CDSCO** • NDCT Rules 2019, Rules 7–8; MDR-2017 refers to the ethics committee in Rule 50
-  **CLA permission, Form MD-22 → MD-23** • Seventh Schedule: consent form, investigation plan, insurance; start within one year
-  **CTRI registration** • prospectively, before the first participant is enrolled; pre-specified outcomes

Three special cases

-  **Dependent relationship • Helsinki 2024, para 27**
Consent must be sought by an appropriately qualified person independent of that relationship; disclose the inventor's interest (CIOMS 2016, Guideline 25).
-  **Vulnerable participants • ICMR 2017, s. 6**
Fluctuating cognition, old age, poverty or dependence on the clinic: extra ethics-committee scrutiny, LAR consent with assent, caregivers involved.
-  **Filming and recordings • separate consent**
Specific, revocable, in writing for public media; a gait video identifies a person; never a condition of care or of study entry (UK GMC as comparator).

Disclose, then manage



THE RULES, IN INDIA AND ABROAD



India

UCMPMD 2024 (amended 30 Apr 2026)
Voluntary code for device companies (DoP circular, 6 Sep 2024): no gifts, travel or hospitality for health professionals; claims accurate and capable of substantiation; complaints to ECMPMD within 90 days

IMC 2002 Regulations, clause 6.8
Doctors: no gifts, travel or hospitality from the pharmaceutical and allied health industry (inserted 10 Dec 2009); the NMC 2023 regulations have been in abeyance since 23 Aug 2023

ICMR 2017 • NDCT Rules 2019
Interests disclosed to the ethics committee and in the consent conversation; an ethics committee registered with CDSCO reviews the investigation plan



Abroad

USA • Open Payments (Sunshine Act)
Manufacturers of drugs and devices report consulting fees, travel, research payments, royalties and ownership interests to physicians every year; CMS publishes them

USA • Harvard Medical School policy, 2010
From disclosure to limits: faculty barred from industry speakers' bureaus; the model for 'disclose, then manage' in this session

UK • Cumberlege Review, 2020
First Do No Harm recommended an expanded GMC register of all doctors' financial and non-pecuniary interests, a Patient Safety Commissioner and device registries

None of these replaces project-level management: independent consent and assessment, a registered protocol, full publication.

03

PART 3 • PERMISSION IN INDIA

Who allows what, for which device?

One Act, one set of Rules, four risk classes. How a wearable like FoGO moves from a test licence to a State manufacturing licence, and what never stops afterwards.



Four risk classes and the First Schedule rules that decide them



Which licence, from which authority, with which form and fee



What a licence holder must keep doing: PSURs, MvPI reports, change control



FoGO ankle module under IEC 60601-1-2 EMC pre-compliance testing, July 2026

India's regulatory architecture

Drugs and Cosmetics Act, 1940 • s.3(b): a medical device is regulated as a 'drug' • all devices notified from 1 Apr 2020 (S.O. 648(E))

Medical Devices Rules, 2017 • G.S.R. 78(E), 31 Jan 2017 • in force 1 Jan 2018 • Rule 4: Class A, B, C, D by the First Schedule



Central Licensing Authority

CDSCO, headed by the DCGI, New Delhi; Medical Device and Diagnostics Division

- Manufacture Class C and D: MD-7 → MD-9
- Import, every class: MD-14 → MD-15
- Test licences MD-12 → MD-13; clinical investigation MD-22 → MD-23
- No predicate in India: MD-26 → MD-27 (Rule 63); Medical Device Officers inspect C/D sites



State Licensing Authority

State or UT Drugs Controller, one per State

- Manufacture Class A and B: MD-3 → MD-5 (loan licence MD-4 → MD-6)
- Relies on a registered notified body's QMS audit of the site (Rule 13)
- Sale, stocking, distribution: MD-41 → MD-42 (Rule 87A, since 30 Sep 2022)
- Class A non-sterile, non-measuring: online registration instead of a licence (G.S.R. 777(E), 2022)

WHO ELSE IS IN THE ROOM



Notified bodies

- Audit Class A/B sites for the Fifth Schedule QMS (Rule 13)
- NABCB-accredited; e.g. TUV Rheinland, Intertek



Testing labs

- Rule 19 Central Medical Device Testing Labs; NABL labs
- AMTZ: EMC, electrical safety, biomaterials



Ethics committees • CTRI

- EC registered with CDSCO approves the study
- CTRI registration before the first participant



MvPI • IPC Ghaziabad

- Device adverse-event reports since 6 Jul 2015
- MDAE form; helpline 1800 180 3024



BIS

- IS 23485 = ISO 13485 + Essential Principles
- ~1,200 device standards; 214 critical devices



NPPA

- DPCO 2013 ceiling prices: stents, knee implants
- Trade-margin caps on five home devices (2021)

From a handful of notified devices to all of them



2026: THREE DOCUMENTS TO READ BEFORE YOU FILE

MDSW guidance 21 Jul 2026

CDSCO/MD/GD/MDSW/01/2026, 62 pages: SaMD, AI/ML, cybersecurity, change control, post-market

G.S.R. 743(E) 14 Aug 2026

Sterilisation-site licence number on labels from 14 Feb 2027; Ninth Schedule testing fees

G.S.R. 744(E) 14 Aug 2026

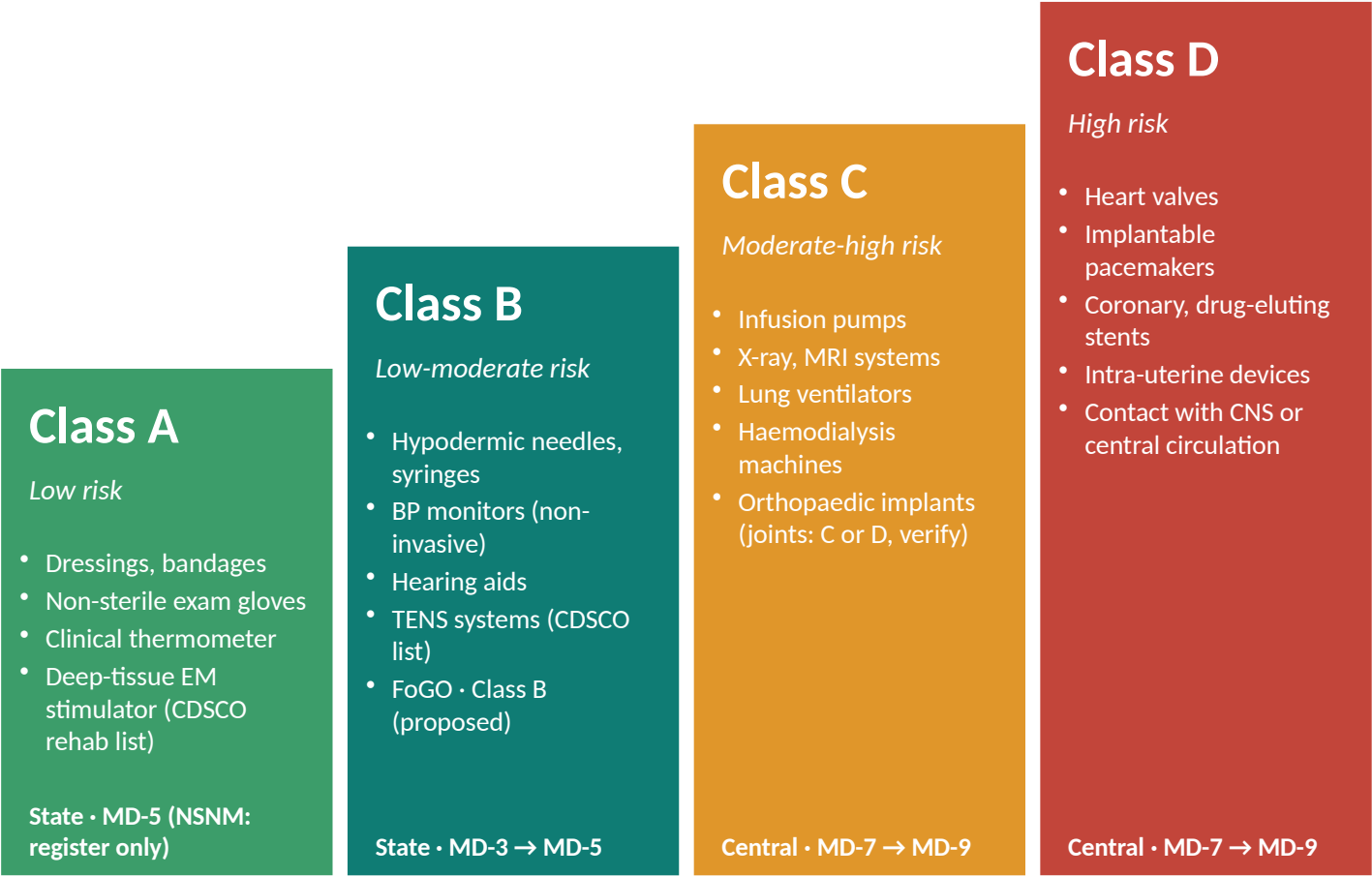
EU approvals added to the Rule 63 trial waiver; Class A QMS self-certification revised

Draft G.S.R. 515(E) 23 Jun 2026

Licence timelines: Class B 140 → 115 days; C/D 105 → 90 days. Not final on 7 Oct 2026



Four risk classes decide the route



FIRST SCHEDULE, PART I: FOUR RULE FAMILIES

- 1 Non-invasive**
Intact skin only → A; channels or stores blood or body fluids → B
- 2 Invasive**
Body orifice vs surgical; transient, short-term, long-term; implants → C or D
- 3 Active**
Therapeutic energy → B; hazardous energy or monitoring vital processes → C
- 4 Special rules**
Medicinal substance → D; non-viable animal tissue → D; IVDs under Part II

WHERE OUR TWO DEVICES WOULD SIT

Class B (proposed)

FoGO: active wearable plus app that cues. The developer's proposal, not a CDSCO classification.

Class to be confirmed

SwaKnee: CDSCO's rehabilitation list puts 'deep-tissue electromagnetic stimulation' in Class A; the active-therapeutic rule suggests B.

Increasing risk → more evidence, higher authority, closer scrutiny

Which licence, from which authority, with which form

Activity	Class A • non-sterile, non-measuring	Class A sterile or measuring • Class B	Class C • Class D	Who decides
Manufacture	Online registration + Essential Principles self-certification (G.S.R. 777(E), 14 Oct 2022)	MD-3 → MD-5	MD-7 → MD-9	State (A/B) • Central (C/D)
Loan licence	—	MD-4 → MD-6	MD-8 → MD-10	State (A/B) • Central (C/D)
Import	MD-14 → MD-15	MD-14 → MD-15	MD-14 → MD-15	Central, every class
Test licence: make or import for study, demo, training	MD-12 → MD-13 • MD-16 → MD-17	MD-12 → MD-13 • MD-16 → MD-17	MD-12 → MD-13 • MD-16 → MD-17	Central
Clinical investigation	MD-22 → MD-23	MD-22 → MD-23	MD-22 → MD-23	Central + registered ethics committee
No predicate in India (Rule 63)	MD-26 → MD-27	MD-26 → MD-27	MD-26 → MD-27	Central
Sale, stocking, distribution	MD-41 → MD-42 (Rule 87A)	MD-41 → MD-42 (Rule 87A)	MD-41 → MD-42 (Rule 87A)	State

A/B licence: ₹5,000/site + ₹500/device

C/D licence: ₹50,000/site + ₹1,000/device

Import: US\$1,000–3,000/site + \$50–1,500/device

MD-13: ₹500/device

MD-41/42: ₹3,000

Teal = State Licensing Authority • Indigo = Central (CDSCO). Licences are perpetual: pay the retention fee (same amount) every 5 years; late fee 2% a month; a manufacturing licence lapses after 180 days unpaid. Clinical-investigation fee: see the Second Schedule (verify).

The route to a manufacturing licence



Class A is granted on self-certification and audited after; Class B is audited before grant; Class C and D are inspected by CDSCO. Higher class, closer scrutiny.

Importing, registering and relying on foreign approvals



Import: Central Licensing Authority, every class

Rules 34–36, MDR-2017 • Form MD-14 → licence MD-15

1

Appoint an Indian authorised agent

The agent or importer files on the Medical Device online system (cdscomonline.gov.in)

2

Apply in Form MD-14 to CDSCO

One route for Class A to D; the overseas site may be inspected

3

Pay Second Schedule fees in US dollars

A: \$1,000/site + \$50/device • B: \$2,000 + \$1,000 • C/D: \$3,000 + \$1,500

4

Decision within 9 months

Practitioners quote 6–9 months; MD-15 is perpetual with a 5-yearly retention fee (90-day grace)

5

Report what happens abroad

Rule 38: withdrawal, restriction or cancellation elsewhere → CDSCO within 15 days. Test imports: MD-16 → MD-17, \$100/device



Class A non-sterile, non-measuring: register, don't license

G.S.R. 777(E), 14 Oct 2022 (Sixth Amendment)

- Upload on the online system: site name and address; device details; an undertaking that the device is Class A, non-sterile, non-measuring; a self-certified Essential Principles checklist
- Registration number goes on the label; 'Reg. No.' wording proposed in the Dec 2025 draft
- QMS self-certification revised by G.S.R. 744(E), Aug 2026. Sterile or measuring Class A still needs MD-5



Reliance: no predicate in India, approved abroad

Rule 63, Chapter VIII • Form MD-26 → permission MD-27

- CDSCO may waive the local clinical investigation if the device is approved in the US, UK, Australia, Canada, Japan or, since 14 Aug 2026 (G.S.R. 744(E)), the EU
- Conditions: marketed there for at least 2 years; CDSCO satisfied with safety, performance and post-market data
- A post-marketing clinical investigation in India is still required



recognised regulators

About 70% of what India uses is imported (Parliamentary Standing Committee, Mar 2024). Every importer is a licence holder with the same post-market duties as a manufacturer.

Eight duties that never stop



Essential Principles

Rule 6 • EP checklist

- Checklist filed with every application
- Class A non-sterile, non-measuring: self-certified



Quality management system

Fifth Schedule • IS 23485

- ISO 13485-aligned manual, procedures, records
- Notified-body audit (A/B) or CDSCO inspection (C/D)



Labelling

Rule 44 • Fourth Schedule

- Name, manufacturer, net quantity, month and year of manufacture and expiry, licence number
- From 14 Feb 2027: sterilisation-site licence number (S.M.L.)



Unique Device Identification

Rule 46 • date deferred

- Fixed start date removed by G.S.R. 918(E), 31 Dec 2021
- No date fixed in 2026: plan the UDI anyway



Your licence

MD-5 or MD-9 • perpetual

- Retention fee every 5 years, same amount
- Rule 43A: suspension or cancellation after show-cause for non-compliance



Standards

Rule 7 • BIS, then ISO/IEC

- BIS: ~1,200 device standards; 214 critical devices prioritised
- Testing fees now in a Ninth Schedule (G.S.R. 743(E))



Records and traceability

Chapter IIIA • Rule 87A

- Batch or serial number and software version for every unit
- Registration number on the label; sellers hold MD-42



Change control

MDSW guidance, 21 Jul 2026

- Version control, validation, documented change management
- Major change: prior approval; minor: notification (verify)

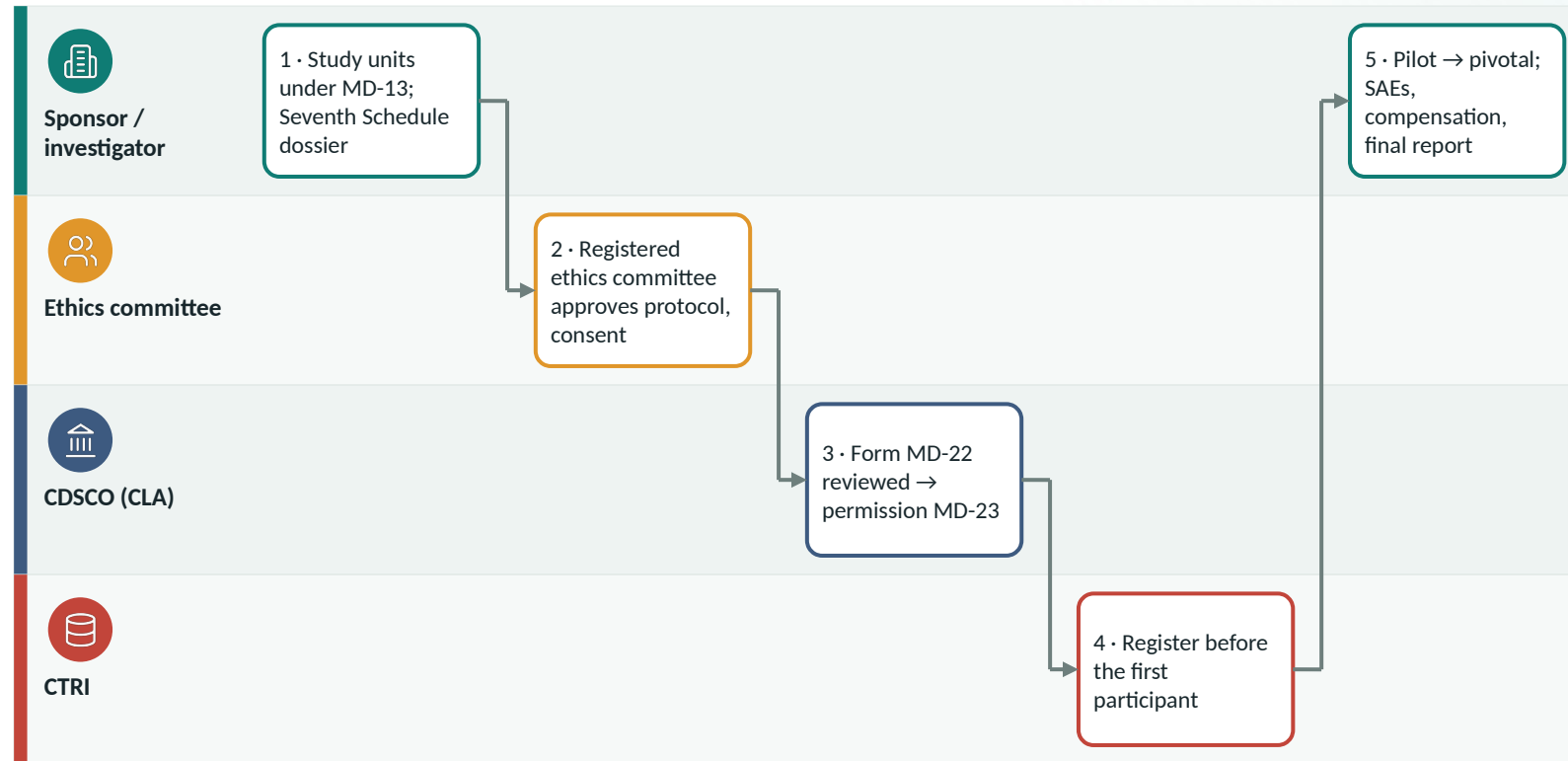


Vigilance

Rule 38 • MvPI circular 15 May 2024

- Serious adverse events: report within 15 days
- Action abroad → CDSCO within 15 days; MDAE form to MvPI

Clinical investigation under MDR-2017



Seventh Schedule dossier

Filed with Form MD-22 (CDSCO checklist)

- Clinical investigation plan
- Investigator's brochure
- Design-control documents
- Design verification and validation report
- Essential Principles checklist
- Informed consent form
- Investigator undertaking
- Ethics committee approval
- Pilot, pivotal or post-marketing study?



Exemptions and shortcuts

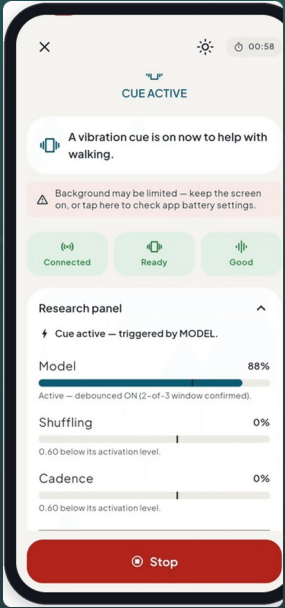
- Rule 63 proviso: approved in the US, UK, Australia, Canada, Japan or (since Aug 2026) the EU, marketed ≥ 2 years → local study may be waived; post-marketing study still due
- Academic study of a licensed device: ethics approval; data not for marketing (verify the Rule text)

Before the first participant: ethics approval, MD-23 permission, CTRI registration, insurance and compensation arrangements. Pilot = exploratory first-in-human; pivotal = confirmatory. Fee waived for government institutions (Rule 51(2)).

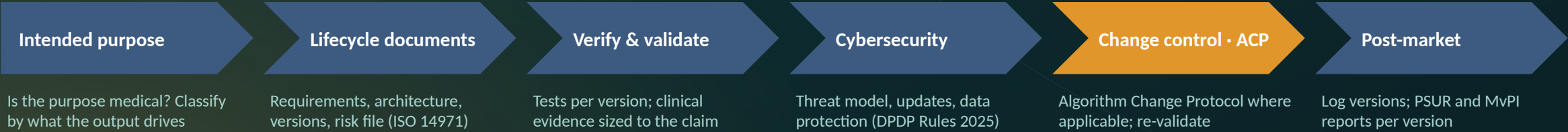
When the software is the device

Severity of the healthcare situation ↓	Significance of the information for the clinical decision →		
	Inform clinical management	Drive clinical management	Treat or diagnose
Non-serious	lowest risk		
Serious		FoGO app • B (proposed)	
Critical			highest risk

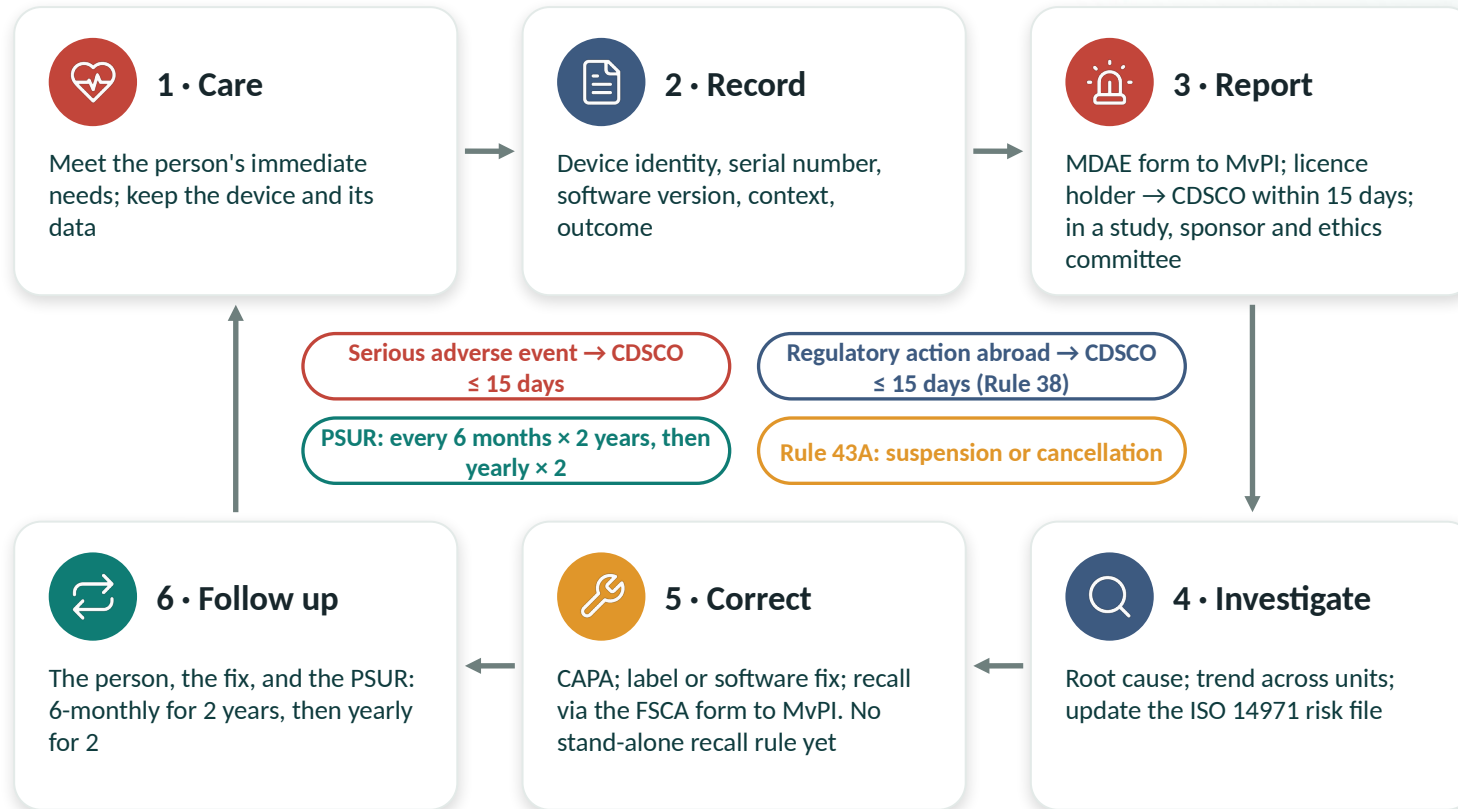
Plus the third criterion: is the medical purpose stated? Illustrative placement of the FoGO app; CDSCO's guidance sets the rule.



- FoGO app: software that drives the device**
- Reads the 100 Hz ankle stream, runs the detection model, confirms a freeze in 2 of 3 windows, then fires the chest cue.
- Software driving a device: classified with the device (verify); FoGO proposes Class B
 - Consultancy reading: retrospective analysis only → A; real-time parameters, no diagnosis → B; informs diagnosis or treatment → C
 - Every result carries a firmware and model version; a threshold change is a design change



Post-market duties and materiovigilance



6 Jul 2015

MvPI launched

IPC Ghaziabad, National Coordination Centre

174

Monitoring centres

hospitals and colleges, ~2022; ~97% of reports

35,391

Reports 2018–24

2026 analysis; 1,931 in 2015–19, 1,277 serious

1800-180-3024

MvPI helpline

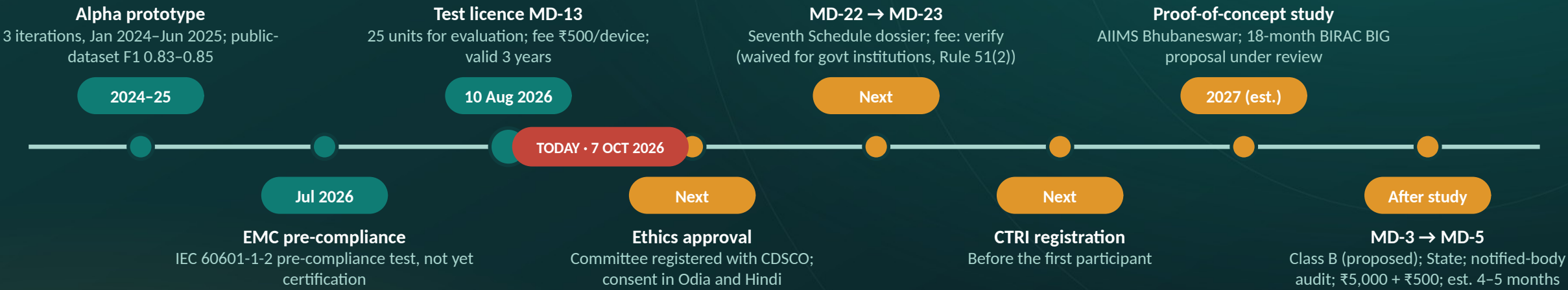
MDAE form v1.2; mvpi-ipc@gov.in

Reporting is becoming a duty

- CDSCO circular 15 May 2024: every licence holder reports suspected unexpected serious adverse events to MvPI
- Mar 2025: expert committee under MvPI; 15 Jul 2025: NMC tells medical colleges to form device adverse-event committees

DePuy ASR hips, 2013 alert: about 4,700 implanted in India, only 882 patients traceable. Traceability starts on day one.

FoGO on the Indian path



Done

Demonstrated

Prototype function, EMC pre-compliance and an MD-13 test licence for 25 units. Units made under MD-13 cannot be sold or used in routine care.

Next 12–18 months

Planned

Ethics approval, MD-23, CTRI, a proof-of-concept study (IDEAL stages 1–2a), then the QMS and an MD-5 application. Not yet funded, approved or permitted.

Estimates, not facts

Estimates

Durations and prices are consultancy or developer estimates: MD-5 in 4–5 months; ISO 13485 in 3–6 months for ₹3–8 lakh; planned price ₹27,000 + ₹3–4,000 a year.

Classify these four devices



Digital thermometer

- Measures body temperature at home
- Battery powered; a display, no app
- Notified as a 'drug' from 1 Jan 2021
- NPPA trade-margin cap since Jul 2021



PEMF knee device

- Pulsed electromagnetic field to the knee
- 45-minute sessions at home (leaflet)
- Applicator plus mains-powered controller
- Leaflet: adjunct for knee osteoarthritis



Wearable + app that cues

- Ankle sensor streams to a phone app
- App detects freezing of gait
- Chest module vibrates a cue
- Holds an MD-13 test licence for 25 units



Coronary stent

- Implanted in a coronary artery
- Stays in the body permanently
- Drug-eluting; NPPA ceiling price
- Notified before 2020: licensed since 2018

THREE QUESTIONS THAT DECIDE IT



Does it enter the body, and for how long? Intact skin, body orifice, surgery; transient, short or long term.



Is it active? Energy delivered, software driving an action, monitoring of vital physiological processes.



What does the label claim? Intended use fixes the class, the authority and the evidence.



Registration only if non-sterile, non-measuring; else State MD-5



State Licensing Authority · MD-3 → MD-5



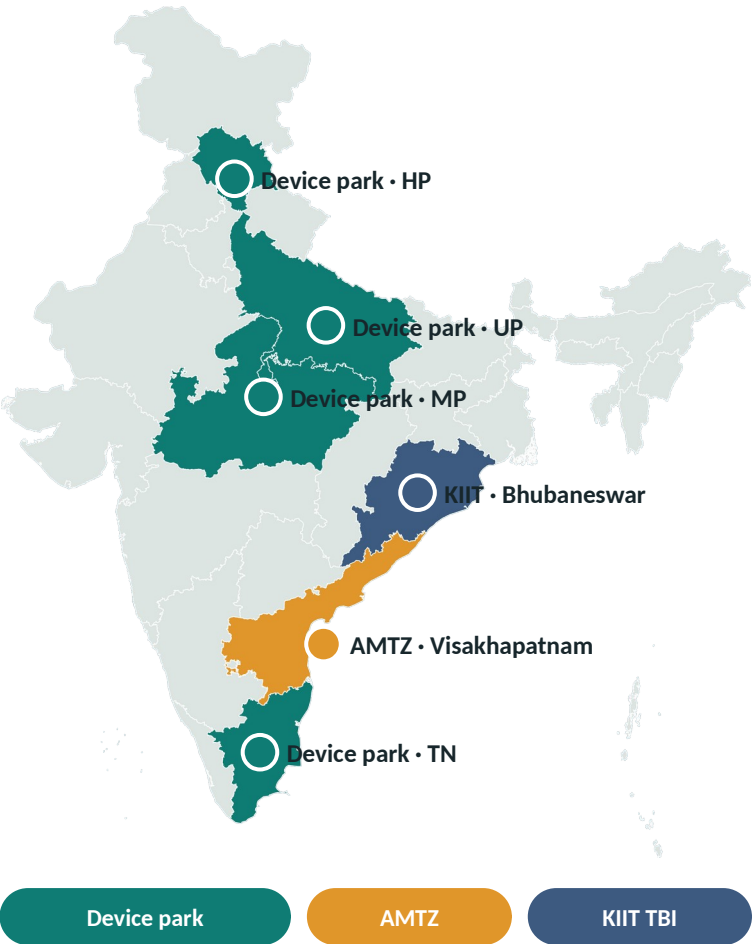
Central (CDSCO) · MD-7 → MD-9



Central (CDSCO) · MD-7 → MD-9

Tables, 90 seconds: write the class, the authority and the form for each device. Watch for the trap: a Class A device can still need a State licence.

Where India builds devices



- 1** **National Medical Devices Policy 2023**
Cabinet, 26 Apr 2023: US\$ 11 bn (2023) → US\$ 50 bn by 2030; six strategies, incl. an export council
- 2** **PLI for medical devices • ₹3,420 crore**
FY2020-21 to FY2027-28; 5% on incremental sales; by Mar 2025: 27 projects, ₹1,153 crore, 54 products
- 3** **Four medical device parks • ₹400 crore**
Up to ₹100 crore each for shared infrastructure: Himachal, Tamil Nadu, Madhya Pradesh, Uttar Pradesh
- 4** **PRIP • ₹5,000 crore • Strengthening scheme • ₹500 crore**
Industry-academia R&D; Nov 2024 scheme: clusters ₹110 cr, clinical studies ₹100 cr, skills ₹100 cr
- 5** **AMTZ Visakhapatnam • EPCMD (22 Sep 2022)**
NABL EMC, electrical-safety and biomaterial labs; 3D printing; KIIT; export council at YEIDA, Noida
- 6** **BIRAC BIG • MedTech Mitra • KIIT TBI**
BIG: up to ₹50 lakh for 18 months; MedTech Mitra (ICMR + CDSCO) hand-holding; FoGO sits at KIIT TBI



04

PART 4 • PERMISSION ABROAD

Same questions, different answers

What is it for, how risky is it, what evidence, who keeps watching? Every regulator asks the same four questions and answers with different forms, fees and clocks.



Who regulates where, and the forum that links them: IMDRF, MDSAP, WHO



Risk classes mapped across India, the USA, the EU, Japan, Canada and Australia



One dossier, many markets: the standards and reliance routes that travel



FoGO ankle module, 2026: one device, classified separately in every jurisdiction

Where the rules come from



● India · CDSCO, IMDRF affiliate since Sept 2024

● EU-27 · MDR 2017/745, ≈48 notified bodies

● IMDRF Management Committee regulators



IMDRF: the regulators' forum

Founded 2011 as successor to the GHTF. 12 Management Committee members (Swissmedic joined March 2025); WHO is an observer. India's CDSCO became an Affiliate Member after the Seattle session, September 2024: working-group access, no vote.



MDSAP: one audit, five regulators

A single ISO 13485-based audit by an authorised organisation satisfies TGA, ANVISA, Health Canada, MHLW/PMDA and FDA. Observers: EU, Singapore HSA, UK MHRA, WHO Prequalification. Mandatory for Canada Class II-IV licences.



WHO: the model and the list

Global Model Regulatory Framework (2017): start with basic controls (law, authority, A-D classes, vigilance, reliance), then expand. WHO-Listed Authorities: 39 agencies after the 7 Aug 2025 designations; first device-regulator list on 23 Jul 2026.

Risk classes across six systems

Risk tier • example	GHTF / IMDRF	India MDR-2017	EU MDR • UK • Australia	Japan PMD Act	Canada	USA FDA
Highest risk • pacemaker	D	D	III	IV	IV	III (PMA)
Moderate-high • infusion pump	C	C	IIb	III	III	II (510(k))
Low-moderate • syringe	B	B	IIa	II	II	II (510(k))
Lowest risk • exam gloves	A	A	I	I	I	I (mostly exempt)
Who decides	Principles only (N77:2012)	SLA for A, B • CLA for C, D	Self-declare I • notified body Is-III	Notification • RCB • MHLW/PMDA	MDEL (I) • MDL (II-IV)	510(k) • De Novo • PMA



Same rules underneath

GHTF/SG1/N77:2012 principles → India First Schedule (Rule 4), EU Annex VIII (22 rules), Japan, Canada and Australia. Four tiers, one logic: invasiveness, duration, energy, body site.



The US is the exception

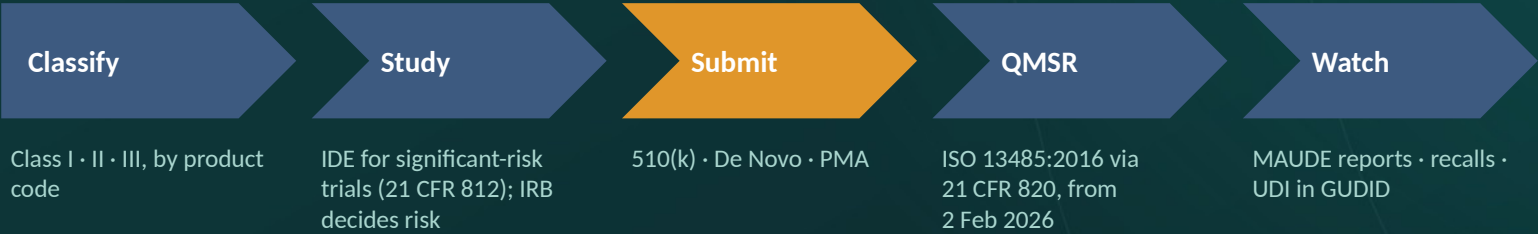
Three classes assigned by product code, not by rules. Class II (510(k)) absorbs most of B and C, so an EU IIb device is often US Class II; some EU III implants are too.



For an Indian team

Classify per jurisdiction: rule wording differs and the class decides the route. A Class B wearable in India maps to EU IIa, Japan II, Canada II; in the US expect Class II and a predicate search.

US FDA: three classes, three doors



**510(k)**

- Class II (and some Class I): substantial equivalence to a predicate
- Fee \$26,067 · small business \$6,517 (FY2026)
- Goal: 95% decided within 90 FDA days
- 3,238 clearances in 2025

**De Novo**

- Novel low-moderate-risk device with no predicate; creates a new classification
- Fee \$173,782 · small business \$43,446
- Later devices can cite the De Novo device as a predicate
- 27 grants reported for 2025

**PMA**

- Class III: reasonable assurance of safety and effectiveness, usually with clinical data
- Fee \$579,272 · small business \$144,818
- Goal: 285 calendar days total time (FY2025–27)
- 41 originals + 2,210 supplements in 2025

**1,246**

Breakthrough designations

2016–2025; 185 reached the market; 136 in 2025

**79 : 1**

510(k)s per original PMA

3,238 clearances vs 41 approvals in 2025

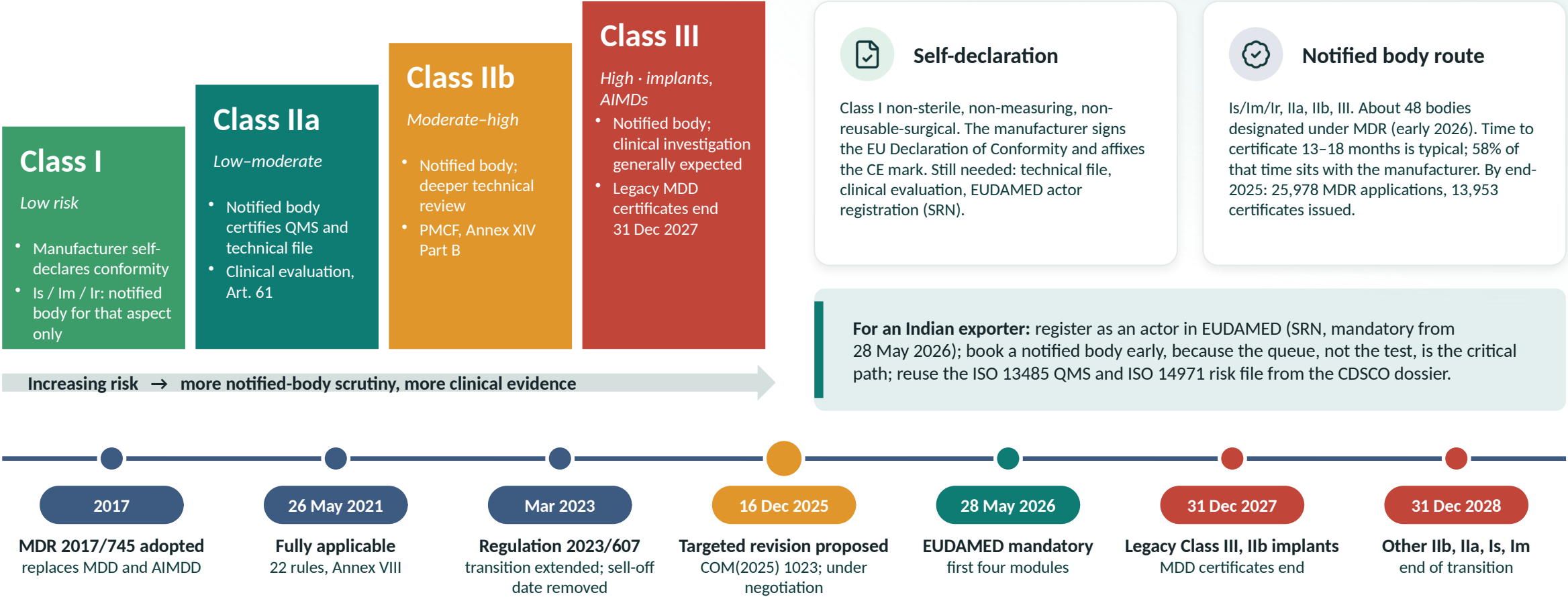
**\$11,423**

Annual establishment registration

No small-business discount; hardship waiver possible

For an Indian exporter: qualify as a small business (≤ US\$100 m gross receipts) and every fee drops to one quarter; build the QMS to ISO 13485:2016 once (India Fifth Schedule, FDA QMSR, MDSAP); run the predicate search before the design freeze.

EU MDR: notified bodies and CE marking



Five more systems in one view



UK
MHRA

CE recognition + reliance

- UK MDR 2002; UKCA via UK Approved Bodies
- CE still accepted: MDD devices to 30 Jun 2028, MDR devices to 30 Jun 2030
- ≈90% of devices on the GB market remain CE-marked
- International reliance route: FDA, Health Canada, TGA, EU approvals
- PMS Regulations in force 16 Jun 2025; Feb 2026 consultation on indefinite CE recognition



Japan
PMDA • MHLW

Classes I-IV, MAH

- PMD Act; GHTF-based Classes I-IV; JMDN codes
- I: notification • II-III: RCB certification where a standard exists • IV: MHLW approval after PMDA review
- Foreign makers act through a Japanese MAH or D-MAH and register as Foreign Manufacturer
- MDSAP member; WHO-Listed Authority since Aug 2025



Australia
TGA

ARTG + reliance

- Therapeutic Goods Act 1989; Classes I, Is, Im, Ir, IIa, IIb, III
- Inclusion in the ARTG through an Australian sponsor
- Accepts comparable-regulator evidence: CE certificates, FDA 510(k)/PMA, Health Canada, Japan, MDSAP (2018), Singapore HSA (2022)
- Australian classification, sponsor and vigilance still apply



China
NMPA

Local testing

- State Council Order 739, in force 1 Jun 2021; MAH regime
- I: filing • II: provincial MPA registration • III: central NMPA; all imports via central NMPA
- II/III: type testing at an NMPA-authorized centre; Class II review target 60 working days
- Clinical evaluation unless on the exemption catalogue or shown equivalent



Canada
Health Canada

Classes I-IV, MDSAP

- Medical Devices Regulations SOR/98-282; Classes I-IV
- I: establishment licence (MDEL) • II-IV: Medical Device Licence (MDL)
- MDSAP certificate mandatory for II-IV: the sole QMS route since 2019
- Since Jan 2026: REP + CESC electronic submissions
- WHO-Listed Authority since Aug 2025

What travels: an ISO 13485 QMS and an MDSAP audit • a CE certificate or FDA decision (TGA and MHRA accept them) • the clinical evaluation and the ISO 14971 risk file.

What stays local: a sponsor or MAH in-country • classification under local rules • registration, labelling and vigilance reporting to that regulator.

India, USA and EU compared



Similar · Lighter · Heavier: presenter's reading

	India • CDSCO	USA • FDA	EU • MDR 2017/745	India is
Law	Drugs and Cosmetics Act 1940; Medical Devices Rules 2017, in force 1 Jan 2018	21 CFR Part 812 (IDE studies) and Part 820 (QMSR); MDUFA user fees	Regulation (EU) 2017/745, amended by 2023/607 and 2024/1860	Similar
Regulator	CDSCO (Central Licensing Authority) for C, D and all imports; State Licensing Authorities for A, B	FDA • CDRH, one federal agency	National competent authorities plus ≈48 notified bodies (NANDO); EMA coordinating role proposed Dec 2025	Similar
Classes	A • B • C • D: Rule 4, First Schedule (GHTF A–D)	I • II • III by product code; II spans GHTF B and C	I (Is, Im, Ir) • IIa • IIb • III: 22 rules, Annex VIII	Similar
Premarket route	A/B: MD-3 → MD-5 (SLA, notified-body audit); C/D: MD-7 → MD-9 (CLA); import: MD-14 → MD-15	Class I mostly exempt; 510(k) • De Novo • PMA; Breakthrough designation	Class I self-declaration; Is to III via a notified body → CE mark with NB number	Lighter
Clinical evidence	Test licence MD-12 → MD-13 for study units; investigation permission MD-22 → MD-23 plus a registered ethics committee	IDE (21 CFR 812) for significant-risk studies, IRB decides; PMA expects clinical data	Clinical evaluation for every device (Art. 61), kept current by PMCF (Annex XIV Part B)	Heavier
QMS standard	Fifth Schedule QMS (ISO 13485-aligned)	QMSR: 21 CFR 820 incorporates ISO 13485:2016, from 2 Feb 2026	ISO 13485:2016; QMS certified by the notified body	Similar
Audit body	NABCB-accredited notified bodies audit A/B sites; CDSCO Medical Device Officers inspect C/D	FDA inspection (Compliance Program 7382.850) or an MDSAP audit	Notified body audit and surveillance; EUDAMED actor registration (SRN)	Similar
UDI	Rule 46 deferred "till further orders" (G.S.R. 918(E), 31 Dec 2021)	UDI on labels, records in GUDID; Class I consumer devices added 2022	UDI and device registration in EUDAMED, mandatory from 28 May 2026	Lighter
Post-market	PSUR six-monthly for 2 years then yearly for 2; MvPI adverse-event reporting (IPC, since 2015)	Adverse events into MAUDE; recall database; UDI expected in recall notices	Vigilance reporting; PMCF keeps the clinical evaluation current; EUDAMED surveillance module	Similar
Time (reported ranges)	A: licence ≤45 days; B: up to 140 days; C/D: 105–150 days; import ≤9 months	510(k): 90 FDA-day goal, 127–159 calendar days historically; PMA: 285-day goal	Certificate in 13–18 months typical; 6–12 months for QMS-only (Team-NB 2025)	Lighter
Cost (reported ranges)	A/B ₹5,000 per site + ₹500 per device; C/D ₹50,000 + ₹1,000; import US\$1,000–3,000 per site + \$50–1,500 per device	510(k) \$26,067 (small \$6,517); De Novo \$173,782; PMA \$579,272; + \$11,423 a year	No public tariff; notified-body quotes vary, reported order of magnitude tens of thousands of euros	Lighter

One dossier, many markets



ISO 13485:2016

Quality management. Reconfirmed 31 Oct 2025; basis of FDA QMSR, MDSAP and India's Fifth Schedule



ISO 14971:2019

Risk management, with ISO/TR 24971:2020 guidance; IEC 60601-1 now references it undated



ISO 14155:2026

Clinical investigation GCP; published 23 Mar 2026, no transition; estimands, CEC and DMC



ISO 10993-1

Biological evaluation inside the risk process; endpoints by contact type and duration



IEC 60601-1 ed. 3.2

Electrical safety and essential performance (2005 + A1:2012 + A2:2020); EMC via 60601-1-2



IEC 62304

Software life cycle (2006 + A1:2015); edition 2 due Aug 2026; two safety classes, AI/ML



IEC 62366-1

Usability engineering (2015 + A1:2020): find use errors before patients do



IEC 81001-5-1:2021

Cybersecurity life cycle; FDA-recognised; EU harmonisation due 27 May 2028

Reliance routes an Indian dossier can use



India • CDSCO

IMDRF affiliate 2024; MDR-2017 technical file, QMS and risk file



FDA • MDSAP member; its decisions count for TGA and MHRA reliance



CE certificate: accepted by TGA; valid in Great Britain to 2028 / 2030



MHLW/PMDA • MDSAP member; WHO-Listed Authority, Aug 2025



TGA • accepts CE, FDA, Canada, Japan, MDSAP and HSA evidence



Health Canada • MDSAP mandatory (II-IV); WHO-Listed Authority

Lessons for Indian innovators: write the technical file once to ISO 13485 and ISO 14971; run the first study to ISO 14155:2026; choose the first foreign market by reliance: TGA and MHRA accept FDA and CE decisions, so one of those two unlocks three markets.

05

PART 5 • ACCESS AND SAFETY

Can they afford it, use it, and who answers later?

Capability audits, the total cost of a course and honest brochures before launch; a named owner for every step after it.



Who can't use it? Seven capabilities, plus the demands India adds



Total cost of use, and how far NPPA's price powers reach under DPCO 2013



Decisions 4 and 5: a brochure, a software update; then the harm loop and MvPI



SwaKnee tablet controller and knee applicator (supplied product photograph; no patient shown)

Who can't use it? Audit the demands a device makes

Illustrative ratings of the demand each task makes on the seven capabilities of the Cambridge EDC Inclusive Design Toolkit

Capability	What FoGO asks of Ramesh	Demand	What SwaKnee asks of Kamala	Demand
Vision	Read the app screen; see the LED and the pairing prompts	Medium	Read the controller display; set the session timer	Medium
Hearing	App alerts only; the cue itself is a vibration	Low	Beeps and error tones from the controller	Low
Thinking	Pair the phone, keep the charging routine, trust a cue mid-freeze	High	Keep a 45-minute daily schedule for 45 days; read error codes	Medium
Communication	Report a fault; consent and instructions in Odia or Hindi	Medium	Follow written steps; ask for help when the cuff slips	Medium
Locomotion	Walk and turn with an ankle module while freezing	High	Travel to the clinic for fitting and follow-up	Medium
Reach and stretch	Bend to the ankle; place the chest patch	High	Bend to wrap the applicator around the knee	High
Dexterity	Strap buckle, adhesive patch, a small charging port	High	Fasten the cuff; connect the cable to the controller	Medium

Pick one demand to redesign first. Classroom simulation: fasten the strap wearing thick gloves. FoGO's team reports consulting 30+ potential users; the common ask was comfort and easier use (project-reported).

India adds demands the toolkit does not list

-  **Language**
Odia, Hindi or English on screen, in the leaflet and at consent
-  **Literacy**
Pictorial steps and teach-back, not a printed manual
-  **Power**
Cuts and voltage swings: charging routines fail first
-  **Smartphone**
Shared, old, or none; no data plan for updates
-  **Caregiver**
Fitting, charging and reporting fall on a relative

Exclusion Calculator: its percentages rest on UK survey data (1996–97); never present them as Indian figures.

Make it testable: IEC 62366-1 usability engineering turns each demand into a requirement; MRCT Accessibility by Design (2023) adds home visits, travel support and accessible consent formats.

Price is only part of the cost

Kamala's 45 days with SwaKnee: six costs, and a question for each



FoGO's planned price

₹27,000

+ ₹3–4,000 a year service plan

Planned, not a market price

Lens from abroad. Cambridge CCHLE: develop → integrate at scale → make affordable. HBS Aravind case: Aurolab lenses made locally under a quality system, so affordability was designed in, not discounted later.

NPPA can reach devices. Devices are 'drugs' under the 1940 Act, so the Drugs (Prices Control) Order 2013 applies.

Para 19: ceiling prices in the public interest	Para 20: MRP monitoring, no rise above 10% in 12 months
Coronary stents capped ₹7,260 bare-metal · ₹29,600 drug-eluting; cuts up to 85% and 74%. Now in Schedule I, revised yearly by WPI 13 Feb 2017	Knee implants, Para 19 Cobalt-chromium primary knee ₹1,58,324 → ₹54,720 (~65%); NPPA found an average trade margin of 313% 16 Aug 2017
Trade margin ≤70% on PTD Pulse oximeters, BP monitors, nebulisers, thermometers, glucometers: 91% of 684 brands cut MRP, by up to 88% 13 Jul 2021	Knee-implant cap runs to Extended year by year since 2017; check NPPA for the next notice. SwaKnee is not an implant: no cap applies to it 15 Nov 2026

Responsible options for an adjunct with modest evidence

- Rent rather than sell; clinic-shared devices
- A trial period with a refund if not tolerated
- State the total cost of a course up front
- Price the service plan, not only the box

Launch with this brochure?

HYPOTHETICAL BROCHURE

Knee relief at home

★ Clinically proven!

★ Doctor recommended

★ Regrows cartilage

★ Avoid surgery forever

Kamala's son found it online. Invented for teaching; not actual SwaKnee material.

The pressures

- The distributor insists on these four claims
- Cash lasts 3 months
- If the company folds, today's users lose service and support

Vote A, B or C. Then each table rewrites one claim in 30 seconds.

- A

Launch as written
The distributor is happy; cash arrives
- B

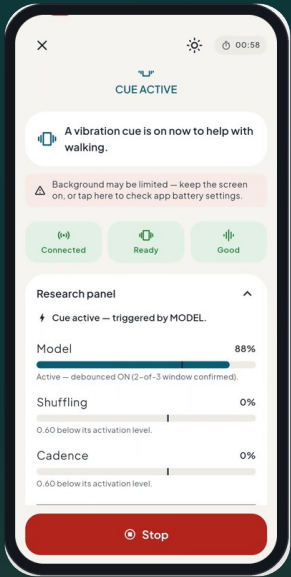
Fix the claims first, then launch
Honest wording this week; keep users supported
- C

Not yet
No launch until independent trials report

The codes behind each answer: UCMPMD 2024 (amended 30 Apr 2026) · DMRA 1954 · ASCI Code I.1 · IMC 2002 cl. 6.8

Brochure claim	Why it fails the Indian codes	What you may say (suggested wording)
“Clinically proven!”	One company study (n = 82), not independently replicated; OARSI 2019 recommends against electromagnetic therapy	“In one company study of 82 adults over 45 days, average pain fell more than with comparison care. Independent trials are needed.”
“Regrows cartilage”	Never measured; no human evidence of cartilage regeneration found	“Intended to ease pain as an adjunct to care. No effect on cartilage has been shown.”
“Doctor recommended”	Implied endorsement: UCMPMD 2024 bars inducements and unsubstantiated claims; IMC 2002 cl. 6.8 binds the doctor	“Ask your doctor whether it suits you. It does not replace the care your doctor advises.”
“Avoid surgery forever”	An absolute cure promise; DMRA 1954 Schedule lists ‘rheumatism’; ASCI Code I.1 demands substantiation	“It does not prevent or replace surgery. If pain persists, see your orthopaedic surgeon.”

Push tonight's FoGO update?



Real prototype app: the cue fired after a debounced 2-of-3 window. Change the rule, change the device.

- Tonight's build (hypothetical update)**
-  Fewer false cues. Today's build fires about 9–11 an hour of walking (0.15–0.18 per minute on public datasets)
 -  May miss more freezes: sensitivity traded for specificity
 -  Tested on stored data only; no user has walked with it

Vote. Then name who in your team can pause or reverse a release.

- A Push to all users tonight**
Everyone gets the new threshold at once
- B Staged release after review**
Shadow mode, a small group, monitoring, rollback trigger
- C Not yet**
Validate with new users first; ask the ethics committee

Evidence belongs to a version: results for version 1 may not cover version 2.

CHANGES NEED GOVERNANCE



CDSCO software guidance, 21 Jul 2026

CDSCO/MD/GD/MDSW/01/2026: lifecycle documentation, version control, verification and validation, documented change management; an Algorithm Change Protocol where applicable. Interprets MDR-2017, adds no new control.



FDA PCCP guidance, Dec 2024 (revised Aug 2025)

Pre-specify the planned modifications, the validation protocol and an impact assessment; changes inside the plan need no new submission. Not a blank cheque: the plan is reviewed first.

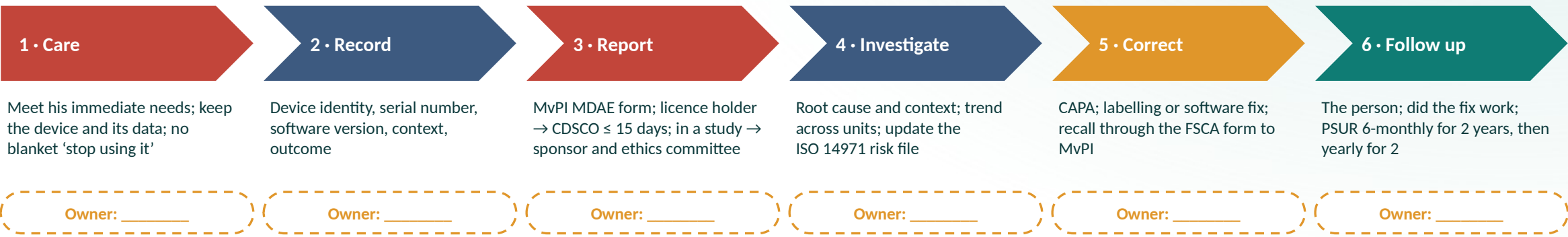


If users are study participants

Notify the ethics committee, seek a protocol amendment, re-consent where the change could affect a participant's decision. Release well: shadow mode → small staged group → pre-set metrics → rollback trigger → tell users and clinicians.

When harm happens, who answers?

Suppose Ramesh falls after a missed cue. Follow the report around the loop and write an owner under each step.



India's reference case: DePuy ASR metal-on-metal hips, recalled worldwide in Aug 2010; CDSCO device alert Dec 2013; import licence cancelled

≈4,700

ASR hips in India

Surgeries 2004–2010; patients had to be found through a helpline

1,080

Traced by Aug 2018

275 had revision surgery (Agarwal committee, 19 Feb 2018)

₹20 lakh

Compensation base

Formula approved 29 Nov 2018; awards ₹30 lakh to ₹1.23 crore by disability and age

Traceability starts on day one: a serial number on every unit, a registry, an in-app 'report a problem' button, a phone line that answers.

Report to MvPI (IPC Ghaziabad, since 6 Jul 2015)

- MDAE form v1.2**
Open to makers, importers, distributors, clinicians and patients; FSMA form for recalls
- 1800 180 3024 • mvpi-ipc@gov.in**
Helpline Mon–Fri 09:00–17:30; 174 monitoring centres (2022 figure)
- CDSCO circular, 15 May 2024**
Every licence holder must report suspected unexpected serious adverse events to MvPI
- Abroad: mostly passive too**
US, EU, Japan and China rely mainly on passive reporting (Kramer 2013); the UK Cumberlege Review (2020) urged a device registry and a Patient Safety Commissioner

Would you let a patient use it now?

The same three facts as the opening vote

-  A working prototype
-  Promising results on public datasets
-  A test licence granted

Four questions to answer silently before you vote

Which patient?	Which use?
Which evidence?	Who is responsible?

Opening	1 = ____	2 = ____	3 = ____
Now	1 = ____	2 = ____	3 = ____

Same three options (one, two or three fingers)

- 1 Routine care**
Suitable for a clinician to recommend to patients now
- 2 Research study**
Suitable for a study with ethics approval and consent
- 3 Not yet**
I need more information before either

A patient-ready decision names

the patient	the purpose	the evidence	the limits	who answers
-------------	-------------	--------------	------------	-------------

Vote again. Then two people whose vote changed: what changed your mind?

Commit to one change in your own project


COMMITMENT CARD • one device, study or project you are working on

Before _____ ,
(next step)

I will _____ .
(action)

Owner: _____
(name)

I proceed only if _____ .
(condition)

-  90 s: write
-  60 s: read it to a neighbour
-  Two examples for the room

Why a card: Stanford Biodesign's Principled Decision-Making brief advises teams to write their ethical principles down early, before the pressured moment arrives. This exercise makes that advice personal; keep the card, or copy it into the printed checklist.

Worked example

"Before our first home test, I will register the study on CTRI. Owner: me. I proceed only if the ethics committee approves and an independent outcome assessor is named."

One example commitment per stage

- Need**
Rewrite the need statement solution-neutral; screen it with ten users before inventing further
- Evidence**
No 'detects' claim until tested on data from Indian patients, with results by subgroup
- People**
Name an independent consent-taker before anyone is approached; consent in Odia or Hindi
- Permission**
Confirm the risk class and hold the MD 13 test licence before a unit leaves the lab
- Access**
Cost the whole course for the household, not the device alone; decide rent or sell
- Safety**
Serial number on every unit; 'report a problem' in the app; name who can pause a release

Patient-impact decision checklist

Stage	The patient asks	Evidence to see	Permission in India (class · form · authority)	Owner	Decision
Need	"Will it help me?"	A solution-neutral need statement; intended use and claim in one sentence. The wording of the claim sets the evidence bar (Stanford Biodesign)	Is it a device? D&C Act s. 3(b)(iv): every device regulated since 1 Apr 2020. Risk class A–D by MDR-2017 Rule 4 and the First Schedule; check CDSCO's classification lists		☐ Go ☐ Not yet
Evidence	"Is it proven, for people like me?"	IDEAL-D stage reached; results by subgroup (MRCT 2020); an ISO 14971 risk file; a version log for software	Prototypes: test licence MD 12 → MD 13 (CLA; ₹500 per device; 3 years). Study: MD 22 → MD 23 with Seventh Schedule documents; CTIRI before the first participant		☐ Go ☐ Not yet
People	"Can I say no? Who gains if I say yes?"	Consent in the participant's language; an independent consent-taker where a patient depends on the researcher (Helsinki 2024, para 27); interests disclosed and managed (ICMR 2017)	Ethics committee registered with CDSCO (NDCT Rules 2019, Rules 7–8; MDR Rule 50). Companies: UCMPMD 2024. Doctors: IMC 2002 cl. 6.8		☐ Go ☐ Not yet
Permission	"Is it allowed for this use?"	QMS to the Fifth Schedule (ISO 13485-aligned); Essential Principles checklist; Rule 44 labelling; software lifecycle and change control (CDSCO MDSW guidance, 21 Jul 2026)	Class A/B: MD 3 → MD 5, State Licensing Authority, notified-body audit. Class C/D: MD 7 → MD 9, CDSCO. Class A non-sterile non-measuring: registration only (G.S.R. 777(E), 2022). Import, any class: MD 14 → MD 15		☐ Go ☐ Not yet
Access	"Can I afford it and use it?"	A capability audit (Cambridge toolkit) plus the Indian demands; the total cost of a course and who pays each step; claims matched to evidence	NPPA under DPCO 2013: Para 19 ceilings (stents and knee implants, 2017), Para 20 MRP monitoring, trade-margin caps (2021). Claims: UCMPMD 2024, DMRA 1954, ASCI Code. Sellers: MD 41 → MD 42 (State)		☐ Go ☐ Not yet
Safety	"Who answers if it fails?"	A named owner for each step of the harm loop; serial numbers and a registry; monitoring and a rollback trigger for every software change	MvPI MDAE form (IPC; 1800 180 3024); licence holder reports serious events to CDSCO within 15 days (circular 15 May 2024); PSUR 6-monthly × 2 years, then yearly × 2; Rule 43A suspension		☐ Go ☐ Not yet

A workshop synthesis, not a substitute for ethics, scientific, regulatory or quality review. Every 'Not yet' needs an owner and a condition for proceeding. A licence, patent, grant or famous partner is not evidence of readiness.

MDR-2017 forms at a glance

State • SLA (teal)

Central • CDSCO, CLA (indigo)

Activity	Class or scope	Apply in	Granted as	Who decides	Rule, schedule or note
Manufacture	Class A, non-sterile and non-measuring	Online registration	Registration number	Online portal	G.S.R. 777(E), 14 Oct 2022: self-certified Essential Principles checklist and undertaking; no licence
Manufacture	Class A sterile or measuring; Class B	MD-3	MD-5	State (SLA)	Rule 20; a Rule 13 notified body audits the site against the Fifth Schedule QMS
Manufacture	Class C; Class D	MD-7	MD-9	Central (CLA)	Rules 21–23; CDSCO Medical Device Officers inspect the site before grant
Loan licence	Class A; Class B	MD-4	MD-6	State (SLA)	Same fee as the manufacturing licence (Second Schedule)
Loan licence	Class C; Class D	MD-8	MD-10	Central (CLA)	MD-10 issued by the CLA since G.S.R. 188(E), 6 Mar 2019
Test licence	Make units for study, test, evaluation, demonstration or training	MD-12	MD-13	Central (CLA)	₹500 per device. FoGO holds MD-13 for 25 units (10 Aug 2026)
Test licence	Import units for the same purposes	MD-16	MD-17	Central (CLA)	US\$100 per device
Clinical investigation	Investigational device, any class	MD-22	MD-23	Central (CLA)	Chapter VII; Seventh Schedule dossier, including the ethics committee approval
New device, no predicate	Import or manufacture under Rule 63	MD-26	MD-27	Central (CLA)	Local study waivable if approved in the US, UK, Australia, Canada, Japan or EU (G.S.R. 744(E), 14 Aug 2026) and marketed 2 years
Import	Any class, through an Indian authorised agent	MD-14	MD-15	Central (CLA)	Rules 34–36; decision within 9 months
Sale and distribution	Any device, including IVDs	MD-41	MD-42	State (SLA)	Rule 87A (G.S.R. 754(E), 30 Sep 2022); ₹3,000; deemed granted unless rejected in 10 days

Also named in the Rules: MD-2 and MD-40 are 'Certificates of Registration' alongside MD-42 (G.S.R. 743(E), 14 Aug 2026); MD-44, a test-report form, is proposed in draft G.S.R. 883(E), 4 Dec 2025. IVD performance-evaluation forms are omitted because their numbers were not verified. Every application is filed on cdscomdonline.gov.in.

Fees and statutory clocks by class (Second Schedule)

Second Schedule fee	Class A • low risk	Class B • low-moderate	Class C • moderate-high	Class D • high
Who licenses manufacture	State Licensing Authority	State Licensing Authority	Central Licensing Authority (CDSCO)	Central Licensing Authority (CDSCO)
Manufacturing or loan licence • MD-5/MD-6 (A, B) • MD-9/MD-10 (C, D)	₹5,000/site + ₹500/device (non-sterile, non-measuring: registration only)	₹5,000/site + ₹500/device	₹50,000/site + ₹1,000/device	₹50,000/site + ₹1,000/device
Retention fee every 5 years (licence is perpetual)	₹5,000/site + ₹500/device	₹5,000/site + ₹500/device	₹50,000/site + ₹1,000/device	₹50,000/site + ₹1,000/device
Import licence • MD-15 (US dollars)	US\$1,000/site + US\$50/device	US\$2,000/site + US\$1,000/device	US\$3,000/site + US\$1,500/device	US\$3,000/site + US\$1,500/device
Test licence • MD-13 (make) • MD-17 (import)	All classes: ₹500 per device to make (MD-13) • US\$100 per device to import (MD-17)			
Sale registration • MD-42 (Rule 87A)	All classes: ₹3,000, with a ₹3,000 retention fee			
Statutory clock	Class A	Class B	Class C	Class D
Manufacturing licence, current rules	Licence on documents within 45 days (no prior audit); notified-body audit within 120 days after grant	Notified-body audit within 90 days, before grant; reported end-to-end: up to 140 days	Inspection within 60 days (Rule 23(1)); decision within 45 days of the report (Rule 21(4)); reported: 105 to 150 days	As Class C
Draft G.S.R. 515(E), 23 Jun 2026 (not final on 7 Oct 2026)	No change proposed	30 + 30 + 20 + 15 + 20 days: 140 → 115 days	Rule 21(4) 45 → 30 days; Rule 23(1) 60 → 55 days: 105 → 90 days	As Class C
Import licence • MD-15; retention fee missed	Import: decision within 9 months of application (6 to 9 months in practice). Retention: late fee 2% a month; licence deemed cancelled after 180 days unpaid (manufacture) or 90 days (import)			

Where the numbers come from: Second Schedule, Medical Devices Rules 2017 (G.S.R. 78(E), 31 Jan 2017). G.S.R. 743(E), 14 Aug 2026, adds a Ninth Schedule of testing fees under Rules 19 and 69 (amounts not captured). Draft G.S.R. 883(E), 4 Dec 2025, restates perpetual validity for MD-2, MD-5, MD-6, MD-9, MD-10 and MD-15. Clinical-investigation (MD-22) fee: verify on cdscomdonline.gov.in before quoting.

CDSCO documents to keep at hand

Document	Number or file	Date	What it covers	Status
Medical Devices Rules, 2017 (consolidated text)	G.S.R. 78(E); 'mdr, 2017' PDF on cdsco.gov.in	31 Jan 2017; in force 1 Jan 2018	Classes A–D (Rule 4, First Schedule); licences by class; Fifth Schedule QMS; Rule 44 labelling; Chapter VII clinical investigation; Rule 63 devices without a predicate	In force
FAQ on Medical Devices Rules, 2017	CDSCO/FAQ/MD/01/2018; IVD FAQ 03-2022; addendum	2018; Mar 2022	Licensing in plain language; Rule 13 notified bodies (NABCB-accredited) audit Class A and B sites; IVD examples: specimen tubes Class A, alcohol analyser Class B	Final
Regulatory pathway under MDR-2017	RegulatoryMDR-2017.pdf	—	One-sheet map of forms and authorities: MD-3/MD-5 and MD-7/MD-9 manufacture, MD-14/MD-15 import, MD-12/MD-13 test licence	Overview
Device classification lists	Public notices of 3 Sep 2020; category-wise lists	3 Sep 2020 onward	About 1,866 devices in 24 categories (Class B alone 779) and 80 IVDs in 3; later: paediatric (23 Aug 2021), cardiovascular revised (Apr 2025), interventional radiology (187)	Living lists
Checklist for Form MD-22	ChecklisFormMD-22MD.pdf	—	Seventh Schedule dossier: investigation plan, investigator's brochure, design V&V, Essential Principles checklist, consent form, ethics approval; pilot, pivotal or post-marketing	Checklist
Guidance on Medical Device Software	CDSCO/MD/GD/MDSW/01/2026	21 Jul 2026 (draft 21 Oct 2025)	62 pages: SaMD, software in devices, cloud and networked systems, AI/ML, IVD software; class by intended use; lifecycle, cybersecurity, AI change planning, post-market duties	Final
Guidance on post-market surveillance	'Draft Guidance on PMS' (2018); CDSCO/IVD/GD/PMS/01/2022	2018; 7 Jul 2022	PMS expectations for licence holders; the IVD draft targets Class C and D and point-of-care or home-use IVDs; neither reported as finalised	Draft
Materiovigilance circular and forms	Circular of 15 May 2024; MvPI MDAE form v1.2; IPC draft IVD-MD AE form	15 May 2024; 17 Feb 2025	Every licence holder reports suspected unexpected serious adverse events to MvPI (IPC; helpline 1800-180-3024); FSCA form for recalls and corrections	Circular
Fee help file; testing laboratories	nmd_fee.pdf (cdscomdonline.gov.in); gazette 2237.pdf	2017; 2018	Second Schedule fees by class; Central Medical Device Testing Laboratories: NIB Noida (IVDs), CDTL Chennai, CDL Kolkata, RDTL Guwahati, CDTL Mumbai	Reference

Amendments to know, by gazette number

G.S.R. 102(E) • 11 Feb 2020
All devices regulated from 1 Apr 2020; Chapter IIIA registration

G.S.R. 754(E) • 30 Sep 2022
Rule 87A: MD-41 → MD-42 for sale, stocking and distribution

G.S.R. 777(E) • 14 Oct 2022
Class A non-sterile non-measuring: registration, no licence

G.S.R. 743(E) • 14 Aug 2026
Rule 44(p) sterilisation-site label from 14 Feb 2027; Ninth Schedule

G.S.R. 744(E) • 14 Aug 2026
EU approvals added to the Rule 63 waiver; Class A QMS self-certification

Drafts • 4 Dec 2025 • 23 Jun 2026
G.S.R. 883(E): perpetual validity, MD-44. G.S.R. 515(E): licence clocks

Evidence to request before stronger claims

Question	FoGO • freezing-of-gait wearable (Class B, proposed)	SwaKnee • PEMF knee system	Benchmark or standard
Outcome	Freeze detection against video-annotated events; then falls, confidence and walking outcomes, not only an F1 score	Pain and function against patient-important thresholds; structural (cartilage) outcomes only if structural claims are made	Daphnet benchmark: 10 patients, 237 video-labelled events, laboratory (2010). Knee-OA pain MCII: -19.9 mm, -40.8% (Tubach 2005)
Comparison	Participant-level validation on unseen users; cue versus no-cue or sham cue	Randomised, blinded, sham-controlled; prospectively registered (CTRI); co-interventions recorded	RESCUE trial: n = 153, home cueing, modest gait gains, falls not counted (2007). Cochrane 2013: 9 trials, 636 adults, pain about 15/100 better than sham
Who and where	Disease stage, walking aids; homes, turns and doorways, not only the laboratory	Age, BMI, severity, comorbidities; rural access and supervision	Wearable reviews: sensitivity 73–100%, specificity 67–100%, mostly laboratory (2017). Knee OA 28.7% in a five-site Indian community survey (2016)
Burden and harm	False cues per hour and latency in real use; missed events, signal loss, skin, data	Adverse events; adherence to 45-minute daily sessions over 45 days; support needs; durability after the course	ISO 14971:2019 risk management; IEC 62366-1 usability engineering; ISO 14155:2026 good clinical practice for device studies
Version	Firmware and model version stated for every result	Controller and applicator version stated for every result	CDSCO software guidance (21 Jul 2026): software lifecycle and AI change planning
Status today	Public datasets only: F1 0.83–0.85 subject-wise, 0.74–0.80 cross-dataset; 0.15–0.18 false alarms per minute; MD-13 test licence for 25 units (10 Aug 2026); AIIMS Bhubaneswar proof-of-concept planned	Company-reported 45-day comparison: n = 40 vs 42, about 32% vs 14% VAS pain reduction; not peer-reviewed; OARSI 2019 and NICE NG226 advise against electrotherapy for OA	IDEAL-D: evaluate devices in stages, with the claim matched to the stage (Sedrakyan 2016)

 FoGO

Reported, preliminary

Detection on public data and a bench cue study; no person with Parkinson's has yet used it at home. The claim today: 'being developed to detect and cue'.

 SwaKnee

Reported, preliminary

One company-reported comparison, not peer-reviewed; OARSI and NICE guidelines advise against the modality. The claim today: 'reported pain reduction in one study'.

 Rule for the Q&A

Before accepting a stronger adjective (validated, proven, safe), ask which row above it rests on, and which device version produced the number.

Teaching materials: Stanford Biodesign and Harvard

[Stanford Biodesign. Our Process](#)

Stanford Mussallem Center for Biodesign. Our Process: Identify, Invent, Implement.

[Stanford Biodesign. Need Statements \(2022\)](#)

Stanford Biodesign. Need Statements. Student Guide to Biodesign; 2022.

[Stanford Biodesign. Principled Decision-Making \(2022\)](#)

Stanford Biodesign. Principled Decision-Making. Student Guide to Biodesign; 2022.

[Chao, Riskin & Krummel \(2010\)](#)

Chao KZ, Riskin DJ, Krummel TM. A patient-centered, ethical approach to medical device innovation. Virtual Mentor (now AMA J Ethics). ...

[Stanford GSB case OIT105 \(2011\)](#)

Burns L, Denend L, Zenios S. The Role of Physicians in Device Innovation: Critical Success Factor or Conflict of Interest? Stanford GSB case OIT105; ...

[Stanford Biodesign in India](#)

Stanford Biodesign. Stanford Biodesign in India (Stanford-India Biodesign, 2007–2014; now School of International Biodesign, AIIMS and IIT Delhi).

[Chaturvedi et al., BMJ Innovations \(2015\)](#)

Chaturvedi J, Logan A, Narayan G, Kuttappa S. A structured process for unmet clinical need analysis for medical device innovation in India: early ...

[Stanford Biodesign. Health Equity](#)

Stanford Biodesign. Health Equity programme and resources.

[HMS Center for Bioethics](#)

Harvard Medical School Center for Bioethics. Fellowship in Bioethics: curriculum.

[MRCT Center. Diversity guidance \(2020\)](#)

MRCT Center of Brigham and Women's Hospital and Harvard. Achieving Diversity, Inclusion, and Equity in Clinical Research: Guidance Document and ...

[MRCT Center. Accessibility by Design \(2023\)](#)

MRCT Center. Accessibility by Design in Clinical Research Toolkit; 2023.

[MRCT Center. Clinical Research Glossary](#)

MRCT Center. Clinical Research Glossary (plain-language definitions; v3.0, 2025).

[MRCT Center. post-trial access for devices \(2025\)](#)

MRCT Center Post-trial Continued Access Task Force. Post-trial, Continued Access Responsibilities to Investigational Significant-Risk Devices: ...

[MRCT Center. Return of Aggregate Results \(2017\)](#)

MRCT Center. Return of Aggregate Results to Participants: Guidance Document and Toolkit, v3.1; 2017.

[Cohen, Gerke & Kramer, Milbank Q \(2020\)](#)

Cohen IG, Gerke S, Kramer DB. Ethical and legal implications of remote monitoring of medical devices. Milbank Q. 2020;98(4):1257–1289.

[Petrie-Flom, Diagnosing in the Home](#)

Petrie-Flom Center, Harvard Law School. Diagnosing in the Home: the ethical, legal and regulatory challenges of digital home health (2021–2024).

[Gerke, Babic, Evgeniou & Cohen \(2020\)](#)

Gerke S, Babic B, Evgeniou T, Cohen IG. The need for a system view to regulate artificial intelligence/machine learning-based software as medical ...

[Simon, Shachar & Cohen, JAMA \(2022\)](#)

Simon DA, Shachar C, Cohen IG. Unsettled liability issues for 'prediagnostic' wearables and health-related products. JAMA. 2022;328:1391.

[Kramer et al., PLoS Med \(2013\)](#)

Kramer DB, Tan YT, Sato C, Kesselheim AS. Postmarket surveillance of medical devices: a comparison of strategies in the US, EU, Japan, and China. ...

[HMS conflict-of-interest policy \(2010\)](#)

Harvard Medical School. Policy on Conflicts of Interest and Commitment (2010 revision).

[HBS case, Aravind Eye Hospital \(1993\) • no public link](#)

Rangan VK. The Aravind Eye Hospital, Madurai, India: In Service for Sight. Harvard Business School case 593-098; 1993.

Teaching materials: Cambridge, and evaluation frameworks

[Cambridge EDC, Inclusive Design Toolkit](#)

University of Cambridge Engineering Design Centre. Inclusive Design Toolkit and Exclusion Calculator (Clarkson, Waller, Coleman).

[Engineering Better Care \(2017\)](#)

Royal Academy of Engineering, Academy of Medical Sciences, Royal College of Physicians. Engineering Better Care: a systems approach to health and ...

[Cambridge Engineering module 4G9](#)

University of Cambridge Department of Engineering. Part IIB module 4G9: Biomedical Engineering / Biomedical Innovation (syllabus).

[Cambridge Institute for Biomedical Innovation \(2026\)](#)

University of Cambridge. New Cambridge institute will support high-value medical device translation; January 2026.

[Cambridge Judge, CCHLE](#)

Cambridge Judge Business School. Centre for Health Leadership and Enterprise.

[University of Cambridge COI policy](#)

University of Cambridge. Conflict of Interest Policy.

[IE Cambridge, How to Guide: Spinouts \(2024\)](#)

University of Cambridge Innovation and Entrepreneurship. How to Guide 3: Spinouts; 2024.

India: regulation, guidance, pricing and safety (1/7)

[MoHFW: CDSCO becomes IMDRF Affiliate Member \(3 Oct 2024\)](#)

Ministry of Health and Family Welfare. Press release: CDSCO becomes Affiliate Member of the International Medical Device Regulators Forum (approved ...

[CDSCO classification of medical devices \(Classes A to D\)](#)

Central Drugs Standard Control Organisation. Classification of medical devices under the Medical Devices Rules, 2017: Class A low risk, B ...

[CDSCO medical device portal fee schedule \(Second Schedule\)](#)

CDSCO Medical Device online portal. Fee help file (Second Schedule, MDR-2017): manufacturing licence Class A/B Rs 5,000 per site + Rs 500 per device; ...

[Lawrbit, medical device import licence \(MD-14/MD-15\) fees](#)

Lawrbit. Medical device import licence India procedure: Form MD-14 to MD-15; fees US\$1,000 (A), \$2,000 (B), \$3,000 (C/D) per site plus \$50 / \$1,000 / ...

[AZB Partners, Medical Devices Rules 2017 summary](#)

AZB Partners. Medical Devices Rules, 2017: State Licensing Authority for Class A and B manufacture; Central Licensing Authority for Class C and D and ...

[Applied Clinical Trials, new regulations for medical devices in India \(2017\)](#)

Applied Clinical Trials. New regulations for medical devices in India: Class A licence within 45 days with audit within 120 days after grant; Class B ...

[S.S. Rana, faster manufacturing licence approvals \(draft 2026 amendment\)](#)

S.S. Rana & Co. Medical Devices Rules: faster manufacturing licence approvals (draft G.S.R. 515(E), 23 June 2026; current Class C/D timeline up to ...

[Corpseed, import licence Form MD-14 and MD-15](#)

Corpseed. Permission to import medical devices, Form MD-14 and Form MD-15: Central Licensing Authority decides within 9 months; 6 to 9 months in ...

[SCC Online, Medical Devices \(Amendment\) Rules 2021 and UDI \(Rule 46\)](#)

SCC Online. Draft Medical Devices (Amendment) Rules 2021 introduces unique device identification (Rule 46; final G.S.R. 918(E), 31 December 2021 ...

[Navigator Global, India registration and post-market compliance](#)

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Regulation abroad, ethics and standards (1/2)

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